

# Stopped Information Complexity for Stochastic Bandit Convex Optimization

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## Abstract

We introduce *stopped information complexity* for stationary stochastic bandit convex optimization, a local cost-to-progress ratio optimized over variable-length, observation-adaptive experiments. Its cost is scale-weighted query count plus exploratory regret; its progress is information about one retained near-optimal action or the first constant-factor reduction in posterior risk.

The main result is a two-sided transfer: a uniform stopped ratio  $\Lambda$  yields  $\tilde{O}(\sqrt{d(1+\Lambda)T})$  regret, while a posterior state at risk scale  $b$  with stopped value  $\lambda$  forces  $\Omega(\sqrt{d\lambda n})$  regret at the calibrated horizon  $n \asymp d\lambda/b^2$ . At a fixed state, the ratio is an information-penalized Bellman stopping problem. The upper proof glues local solutions by retaining one target across risk scales and amortizing information and risk reduction; the converse terminalizes a low-regret policy at its first risk reduction. Measurable implementation and Gaussian Bayes–minimax duality close the remaining quantifiers. Writing  $\beta_{\mathfrak{F}}$  for the polynomial growth exponent in  $d$  of minimax regret normalized by  $\sqrt{T}$ , uniformly over polynomial horizons, and  $\alpha_{\text{stop}}$  for the corresponding exponent of the stopped profile over inverse-polynomial risk scales, we obtain the exact identity

$$\beta_{\mathfrak{F}} = \frac{1 + \alpha_{\text{stop}}}{2}.$$

For the dimple family of Bakhtiari, Lattimore, and Szepesvári, one-query and fresh-uniform-direction ratios are  $\Omega(d^2)$ , whereas an unrestricted stopped experiment attains  $\tilde{O}(d)$ ; adaptive continuation removes the apparent one-step obstruction. Thus the dimension of a one-query local problem need not be the dimension of its observation-adaptive continuation. Determining the stopped profile for the full convex class remains equivalent, at the exponent level, to closing the minimax dimension gap.

## 1 Introduction

Let  $C \subset \mathbb{R}^d$  be a convex body. A continuous convex loss  $F : C \rightarrow [0, 1]$  is drawn once and remains fixed. At round  $t$ , a learner chooses  $X_t \in C$  and observes

$$O_t = F(X_t) + \xi_t, \quad \xi_t \stackrel{\text{iid}}{\sim} N(0, 1).$$

Its cumulative regret is

$$R_T = \sum_{t=1}^T \{F(X_t) - \min_{x \in C} F(x)\}.$$

This is stationary stochastic bandit convex optimization (BCO), equivalently noisy zeroth-order convex optimization with cumulative rather than terminal loss. In the nontrivial long-horizon

regime, the best general upper bound is  $\tilde{O}(d^{3/2}\sqrt{T})$  (Lattimore and György, 2023; Fokkema et al., 2024). Rajaraman’s Corollary 1.2 gives  $\Omega(d^{5/4}\sqrt{T/\log(ed)})$  regret on the Euclidean ball when  $T \gtrsim d^3/\log(ed)$  (Rajaraman, 2026). This is the first general lower bound with dimension dependence strictly larger than the  $\Omega(d\sqrt{T})$  inherited from linear bandits (Dani et al., 2008; Shamir, 2013). The numerical gap is therefore narrow but conceptually unresolved: what feature of a convex function costs more than learning an unknown linear direction? Neither endpoint is known to be sharp. Throughout, the *full bounded convex class* consists of all continuous  $[0, 1]$ -valued convex losses on  $C$ .

**Why one query can be too small.** The information ratio compares squared conditional regret with the information in one query (Russo and Van Roy, 2016, 2018; Lattimore and György, 2021b), while the decision–estimation coefficient (DEC) balances immediate loss against statistical separation in one interaction (Foster et al., 2021, 2022, 2023). The algorithmic information ratio (AIR) gives an exact posterior-coordinate regret identity whose logarithmic potential telescopes across rounds (Xu and Zeevi, 2025). These methods are sequential, but their local optimization is over one query–observation pair and is recomputed after each observation. This distinction matters in BCO: one observation may reveal the geometry in which the next probe should be taken. Indeed, Bakhtiari et al. (2025) construct a smooth convex dimple family whose one-query coefficient is of order  $d^2$ . Such a calculation need not remain hard when later directions can depend on earlier observations.

Interactive Fano inequalities already compare the laws of complete adaptive transcripts and are therefore multistep from the outset (Chen et al., 2024), but they are lower-bound tools. Two converse steps should be distinguished. Interactive Fano or Assouad can certify that every stopped experiment at a given state is costly. Our lower transfer is different: conditional on such a local certificate, it converts any low-regret policy into a successful first-hit experiment and invokes the defining ratio. The linear calibration below uses a stopped-transcript Assouad argument.

On the upper side, the posterior-coordinate AIR identity motivates the fixed-index information potential (Xu and Zeevi, 2025). Its posterior-averaged drift is mutual information, and the Bellman-sufficient framework of Xu (2026a) supplies the continuation recursion. For a two-sided characterization in BCO, we therefore take the local unit to be a stopped experiment whose query directions, duration, and terminal action may all depend on the observations collected within the experiment.

Let  $r_B(h)$  be posterior Bayes risk after history  $h$ , and fix a scale  $b$  with  $r_B(h) \asymp b$ . For the analysis, draw a near-optimal action  $A_b$  once and retain it across all later experiments at scale  $b$ . For an adaptive experiment stopped no later than the first history at which  $r_B \leq \gamma b$ , let  $Z$  be its complete transcript,  $N_Z$  its number of queries,  $R_Z$  its exploratory regret, and  $\mathbf{M}$  the event that the smaller risk scale is reached. The local quantity studied in this paper is

$$\Pi_b(h; A_b) = \inf_Z \frac{b^2 \mathbb{E}[N_Z \mid h] + b \mathbb{E}[R_Z \mid h]}{I(A_b; Z \mid h) + d\mathbb{P}(\mathbf{M} \mid h)}. \quad (1)$$

The numerator charges samples and exploratory loss. The denominator records two forms of progress: information about a target retained across later experiments, and the event of reaching a smaller posterior-risk scale. Both terms are necessary. Information alone need not pay for a block that already localizes the optimizer, while risk reduction alone does not telescope across blocks that remain at the same scale.

**The two-sided principle.** For an active posterior state  $s$  and a first-hit experiment  $\mathcal{E}$ , write  $\rho(s, \mathcal{E})$  for the ratio in (1). The mathematical core is the pair of finite-value reductions

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, T) \leq \tilde{O}\left(\sqrt{d\{1 + \Lambda^{\text{policy}}(d, T)\}T}\right), \quad (2)$$

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, n) \geq c_{\gamma}\sqrt{d\lambda n}, \quad n \asymp \frac{d\lambda}{b^2}, \quad (3)$$

whenever one active state at scale  $b$  has stopped value at least  $\lambda$ . The upper reduction is not a formal concatenation of variable-length experiments: it retains one target through repeated visits to each scale, telescopes its information across stopped blocks, and controls returns after a risk reduction. The converse is shorter but equally structural. It turns any low-regret policy into a first-hit experiment by convexly averaging its actions and stopping when posterior risk first falls. This reverse reduction is what makes the stopped ratio necessary rather than merely a sufficient upper-bound certificate.

The two directions initially have different quantifier orders:

$$\underbrace{\sup_s \inf_{\mathcal{E}} \rho(s, \mathcal{E})}_{\text{statewise profile } \Lambda^{\text{state}}} \leq \underbrace{\inf_{\sigma \text{ measurable}} \sup_s \rho(s, \sigma(s))}_{\text{policy profile } \Lambda^{\text{policy}}}. \quad (4)$$

This is not a game-theoretic minimax gap:  $\sigma$  may choose a different experiment at every state. The remaining issue is only whether those statewise choices admit one measurable implementation. We prove that they do for the continuous Gaussian model, without a common deterministic query cap. A direct finite-horizon Bayes–minimax argument also gives

$$\inf_{\pi} \sup_f \mathbb{E}_f R_T = \sup_Q \inf_{\pi} \mathbb{E}_Q R_T. \quad (5)$$

These two closure results turn (2)–(3) into an exact exponent identity. For every nondegenerate class sequence, meaning  $\liminf_{d \rightarrow \infty} \Lambda(d, 2) > 0$ , the common stopped exponent satisfies

$$\boxed{\beta_{\mathfrak{F}} = \frac{1 + \alpha_{\text{stop}}}{2}}. \quad (6)$$

Here  $\beta_{\mathfrak{F}}$  is the polynomial dimension exponent of minimax regret over polynomial horizons. The statement is exact at the exponent level; it is not a pointwise equality between every finite-horizon regret value and one local ratio.

**Logical hierarchy.** The proof contains substantial regularity work, but its components play different roles. The following table makes the hierarchy explicit.

| ingredient                     | role                                                                                                           | status             |
|--------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------|
| upper stopped transfer         | treats a stopped Bellman program as the local problem and amortizes its solutions across posterior-risk scales | rate-bearing       |
| lower stopped transfer         | converts a local hardness certificate into regret by an online-to-batch first-hit argument                     | rate-bearing       |
| adaptive dimple experiment     | tests whether observation-dependent continuation changes the dimension of the local problem                    | geometric content  |
| measurable implementation      | assembles pointwise near-minimizers into one admissible policy                                                 | quantifier closure |
| Gaussian Bayes–minimax duality | identifies worst-prior Bayes and minimax values at a finite horizon                                            | quantifier closure |

The closure steps are logically necessary for the word “exact,” but do not alter the exponent once the finite-value transfers are established. The rate-bearing content lies in the two transfers and in determining the stopped profile itself.

**Adaptive continuation.** Our second result shows that the stopped value can be polynomially smaller than its one-step restriction. On the dimple family, at  $\varepsilon = d^{-1}$  and  $b \asymp d^{-1}$ ,

$$\Pi_b^{[1]} = \Omega(d^2), \quad \Pi_b^{\text{fresh}} = \Omega(d^2), \quad \Pi_b = \tilde{O}(d). \quad (7)$$

Here  $\Pi_b^{[1]}$  allows one query, while  $\Pi_b^{\text{fresh}}$  allows adaptive radii but uses a fresh uniform direction at each query. The unrestricted experiment first learns a weak direction, amplifies it by tangent finite differences, and then contracts its angular error geometrically. The one-step obstruction therefore loses a factor  $\tilde{\Omega}(d)$  once the experiment may choose later directions from earlier observations. This is a local separation, not an unrestricted minimax lower or upper bound.

**What remains open.** The dimple theorem shows that a large one-query coefficient need not describe the global problem. The exact characterization therefore leaves one central geometric question: determine the stopped profile of the full convex class. Combining the best known stochastic regret bounds with Theorem 2.4 places its exponent between  $3/2$  and  $2$ , but the present results do not select either endpoint. In particular, the currently known  $d^{5/4}$  regret lower exponent is a lower endpoint, not a proposed target rate. Section 3 explains what the paper’s examples do and do not resolve.

**Relation to prior work.** [Lattimore and Szepesvári \(2019\)](#) gave a measure-theoretic minimax theorem for finite-action partial monitoring with general signal spaces. Our direct Gaussian argument handles the continuous action space used here; it is supporting infrastructure rather than the source of the rate characterization. For the stationary stochastic model studied here, [Agarwal et al. \(2013\)](#) first obtained  $\tilde{O}(\text{poly}(d)\sqrt{T})$  regret; later localization and second-order methods progressively improved the dimension dependence ([Lattimore and György, 2021a, 2023](#); [Fokkema et al., 2024](#)). Information ratios and DEC formalize one-round regret–information or decision–estimation comparisons and turn them into sequential bounds by re-optimization, Cauchy–Schwarz, and the information chain rule ([Russo and Van Roy, 2016](#); [Lattimore and György, 2021b](#); [Foster et al., 2021, 2023](#)). AIR contributes the exact posterior-coordinate identity used here ([Xu and Zeevi, 2025](#)); related work connects Bayesian information ratios to frequentist DEC ([Neu et al., 2024](#)). Chained ratios can also organize information across action scales, giving the tight linear-bandit rate with fixed two-query updates ([Gouverneur et al., 2024](#)). Our local optimization instead chooses observation-dependent continuation and stopping.

Interactive Fano compares complete adaptive transcript laws and is therefore multistep from the outset ([Chen et al., 2024](#)). It can certify a local premise  $\Pi_b \geq \lambda$ ; the first-hit converse then converts that premise into regret. In the PAC setting, complementary characterizations also show that adaptive querying can be necessary ([Hanneke and Wang, 2025](#)).

Near-optimal targets appear in rate–distortion and satisficing analyses ([Dong and Van Roy, 2018](#); [Russo and Van Roy, 2022](#); [Arumugam and Van Roy, 2021](#)); related localizations include scale-sensitive ratios and pointwise complexities ([Bubeck and Sellke, 2023](#); [Li and Xu, 2026](#); [Xu, 2026b](#)). Polynomial gains from measurements that orient later probes also arise in best-arm identification ([Maiti et al., 2026](#)). Our recursion specializes the exact-posterior Bellman framework of [Xu \(2026a\)](#) to a retained sublevel-set target. Classical sequential design likewise optimizes adaptive experiments

and stopping (Chernoff, 1959; Naghshvar and Javidi, 2013), but here every query also incurs BCO regret and progress may be a posterior-risk reduction.

Finally, Zhang et al. (2026) obtain fully adaptive lower bounds for noiseless terminal-accuracy convex optimization. Their posterior-stability mechanism is relevant to direct profile lower bounds, but exact queries with no cumulative loss define a different problem from the Gaussian regret model studied here.

Section 2 states the two-sided principle and its exact exponent consequence. Section 3 compares the dimple separation with the full-class profile and isolates the remaining question. Section 4 separates the reusable stationary template from its BCO-specific ingredients. The appendices prove the stochastic transfer and calibrations, the dimple separation, and the measurable and minimax completion steps.

## 2 The two-sided stopped-information principle

Formally, a loss-class sequence  $\mathfrak{F} = (\mathfrak{F}_d(C))$  assigns to each dimension  $d$  and each known compact convex body  $C \subset \mathbb{R}^d$  with nonempty interior a Borel class of continuous  $[0, 1]$ -valued convex losses. Suprema over  $C$  below range over these bodies, and  $\Pi_T(C)$  denotes the randomized nonanticipating Borel policies on  $C$  for horizon  $T$ .

We first state the main result. Fix  $0 < c_A < \gamma < \vartheta < 1$ . At risk scale  $b$ , use the persistent target

$$A_b^\circ \mid (F = f) \sim \text{Unif}\{x \in C : f(x) - \min_C f \leq c_A b\}. \quad (8)$$

Its construction and information bound are proved in Lemma A.1. A state  $s = (Q, b)$  is active when  $\vartheta b < r_B(Q) \leq b$ . An admissible experiment uses a positive, integrable number of queries and stops no later than the first posterior with Bayes risk at most  $\gamma b$ . If  $Z$  is its transcript,  $N_Z$  its length,  $R_Z$  its exploratory regret, and  $M$  the lower-scale hit, put

$$\rho(s, \mathcal{E}) = \frac{b^2 \mathbb{E}_s N_Z + b \mathbb{E}_s R_Z}{I_s(A_b^\circ; Z) + d\mathbb{P}_s(M)}. \quad (9)$$

For a known body  $C$ , let  $S_{\mathfrak{F}, C}(d, B)$  be the active states with  $B^{-1} \leq b \leq 1$ ; it is standard Borel because posterior risk is Borel. Define the *statewise profile* and the *measurable-policy profile* by

$$\Lambda_{\mathfrak{F}, C}^{\text{state}}(d, B) = \sup_{s \in S_{\mathfrak{F}, C}(d, B)} \inf_{\mathcal{E}} \rho(s, \mathcal{E}), \quad (10)$$

$$\Lambda_{\mathfrak{F}, C}^{\text{policy}}(d, B) = \inf_{\sigma_C \text{ universally measurable}} \sup_{s \in S_{\mathfrak{F}, C}(d, B)} \rho(s, \sigma_C(s)), \quad (11)$$

Here  $\sigma_C$  is one universally measurable block policy on the augmented belief state: from the initial state  $s$  and each subsequent local history, it chooses the next action or stops;  $\sigma_C(s)$  denotes the experiment that it induces from  $s$ . Since  $C$  is known, write

$$\Lambda_{\mathfrak{F}}^\diamond(d, B) := \sup_C \Lambda_{\mathfrak{F}, C}^\diamond(d, B), \quad \diamond \in \{\text{state}, \text{policy}\}. \quad (12)$$

Always  $\Lambda^{\text{state}} \leq \Lambda^{\text{policy}}$ .

For  $\diamond \in \{\text{state}, \text{policy}\}$ , define

$$\alpha_{\text{stop}}^\diamond(\mathfrak{F}) = \sup_{m \in \mathbb{N}} \limsup_{d \rightarrow \infty} \frac{\log\{1 + \Lambda_{\mathfrak{F}}^\diamond(d, d^m)\}}{\log d}. \quad (13)$$

Let

$$\mathfrak{R}_{\mathfrak{F}}^{\text{M}}(d, T) := \sup_C \inf_{\pi \in \Pi_T(C)} \sup_{f \in \mathfrak{F}_d(C)} \mathbb{E}_f R_T, \quad (14)$$

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, T) := \sup_C \sup_{Q \text{ on } \mathfrak{F}_d(C)} \inf_{\pi \in \Pi_T(C)} \mathbb{E}_Q R_T, \quad (15)$$

where the expectations include the Gaussian noise and the learner's randomization.

For  $\diamond \in \{\text{B}, \text{M}\}$ , define

$$\beta_{\mathfrak{F}}^{\diamond} := \sup_{M \in \mathbb{N}} \limsup_{d \rightarrow \infty} \sup_{2 \leq T \leq d^M} \frac{\log\{1 + \mathfrak{R}_{\mathfrak{F}}^{\diamond}(d, T)/\sqrt{T}\}}{\log d}. \quad (16)$$

**Theorem 2.1** (Two-sided stopped-complexity transfer). *For every loss-class sequence  $\mathfrak{F}$ , every  $T \geq 2$ , and every  $T^{-1} \leq \eta < 1/2$ , there is  $C = C_{\gamma, \vartheta, c_A} < \infty$  such that*

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, T) \leq \min \left\{ T, C\{1 + \Lambda_{\mathfrak{F}}^{\text{policy}}(d, T)\} \frac{d\{1 + \log(1/\eta)\}}{\eta} + 2T\eta \right\}. \quad (17)$$

In particular,

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, T) \leq \min \left\{ T, \tilde{O} \left( \sqrt{d\{1 + \Lambda_{\mathfrak{F}}^{\text{policy}}(d, T)\}T} \right) \right\}. \quad (18)$$

Conversely, for every  $0 < \lambda < \Lambda_{\mathfrak{F}}^{\text{state}}(d, B)$ , some active initial prior and scale  $B^{-1} \leq b \leq 1$  satisfy, for every  $n \geq 1$  and  $0 < a < \gamma$ ,

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, n) \geq \min \left\{ abn, \left[ \frac{\lambda d(1 - a/\gamma)}{b} - bn \right]_+ \right\}. \quad (19)$$

In particular, whenever  $d\lambda/b^2 \geq 8$ ,

$$n = \left\lfloor \frac{d\lambda}{4b^2} \right\rfloor, \quad (20)$$

$$\mathfrak{R}_{\mathfrak{F}}^{\text{B}}(d, n) \geq c_{\gamma} \sqrt{d\lambda n}. \quad (21)$$

Consequently, if  $\liminf_{d \rightarrow \infty} \Lambda_{\mathfrak{F}}^{\text{state}}(d, 2) > 0$ , then

$$\frac{1 + \alpha_{\text{stop}}^{\text{state}}}{2} \leq \beta_{\mathfrak{F}}^{\text{B}} \leq \frac{1 + \alpha_{\text{stop}}^{\text{policy}}}{2}. \quad (22)$$

The upper direction rests on one scale-amortization bound. Retain one target  $A_j$  across all stopped blocks  $Z_k$  begun at scale  $b_j$ , and let  $p_k = \mathbb{P}(\mathbf{M}_k \mid H_k)$ . Then

$$\mathbb{E} \sum_{k: j(k)=j} \{I(A_j; Z_k \mid H_k) + dp_k\} \leq d \log \frac{1}{c_A b_j} + \frac{d}{1 - \gamma/\vartheta}. \quad (23)$$

The information terms telescope because the same private target is retained. After a successful risk reduction, the posterior loss of the returned action is a nonnegative martingale starting below  $\gamma b_j$ ; returning to the same active bucket requires it to exceed  $\vartheta b_j$ , so optional stopping bounds the return probability by  $\gamma/\vartheta$ . Multiplying (23) by the scale costs  $b_j^{-1}$  and  $b_j^{-2}$  and summing geometrically proves the upper transfer. For the converse, convex averaging turns cumulative regret  $R_n$  into a terminal action of expected loss at most  $R_n/n$ ; stopping the policy at its first posterior-risk reduction then produces an admissible experiment and yields (19). The complete proof, including horizon truncation and monotone-convergence bookkeeping, is in Appendix A.

**Where Bellman and Fano enter.** For a fixed state, scale, multiplier  $\lambda$ , and query cap, the condition  $\Pi_b \leq \lambda$  is exactly the statement that the offset “cost minus  $\lambda$  times progress” has nonpositive value. Its belief-state value satisfies an information-penalized optimal-stopping Bellman recursion; Appendix C uses that recursion to construct measurable near-minimizers. The global upper proof does not solve the original  $T$ -round Bellman equation. It repeatedly invokes these local stopped programs and combines them through persistent-target information telescoping, a return-time martingale bound, and geometric risk-scale balancing. This is an information-theoretic decomposition, not a claim of general computational tractability.

The lower transfer has a different logic. It assumes a local certificate  $\Pi_b \geq \lambda$  and terminalizes any low-regret policy at the first risk reduction. It is not itself an interactive Fano inequality. Interactive Fano, Assouad, or a Bellman supersolution can prove the local premise; the first-hit argument is the passage from that premise to regret.

## 2.1 Closing the quantifiers

Theorem 2.1 is the rate-bearing statement. Two regularity results complete its quantifiers: one identifies Bayes and minimax values, and the other implements the statewise experiments measurably. They are recorded separately so that this technical completion does not obscure the two statistical reductions.

**Proposition 2.2** (Bayes–minimax identity for Gaussian BCO). *For every finite  $d, T$ , every compact convex body  $C$ , and every Borel class of convex losses  $\mathfrak{F}_d(C) \subset \mathcal{C}(C, [0, 1])$ ,*

$$\inf_{\pi \in \Pi_T(C)} \sup_{f \in \mathfrak{F}_d(C)} \mathbb{E}_f R_T = \sup_{Q \text{ on } \mathfrak{F}_d(C)} \inf_{\pi \in \Pi_T(C)} \mathbb{E}_Q R_T. \quad (24)$$

*The supremum on the right is unchanged if it is restricted to finitely supported priors.*

The next result shows that the two profiles in (10)–(11) have the same value in the models studied here.

**Theorem 2.3** (Measurable implementation of statewise experiments). *Suppose  $C$  is compact and  $\mathfrak{F}_d(C)$  is a Borel subset of  $\mathcal{C}(C, [0, 1])$  with the uniform topology. Under Gaussian feedback, for every Borel target kernel into a standard Borel space and every  $B < \infty$ ,*

$$\Lambda_{\mathfrak{F}, C}^{\text{state}}(d, B) = \Lambda_{\mathfrak{F}, C}^{\text{policy}}(d, B). \quad (25)$$

*For every  $\epsilon > 0$  there is a universally measurable policy whose experiment has a finite cap depending on the state and whose ratio is within  $\epsilon$  of the statewise infimum at every active state.*

For a fixed prior and finite horizon, a universally measurable selector in Theorem 2.3 has a Borel version with the same induced strategic law; Lemma C.1 proves this prior-wise Borelization. Thus the selector is admissible in the Bayes value (15), whose policy class was defined using Borel kernels.

The full class of  $[0, 1]$ -valued continuous convex losses is closed in  $\mathcal{C}(C, [0, 1])$ , so Theorem 2.3 applies. Write  $\Lambda$  and  $\alpha_{\text{stop}}$  for the common profile and exponent, and write  $\mathfrak{R}_{\mathfrak{F}}$  and  $\beta_{\mathfrak{F}}$  for the common Bayes–minimax value and dimension exponent.

**Theorem 2.4** (Exact minimax exponent characterization). *Under the hypotheses of Theorem 2.3, if  $\liminf_{d \rightarrow \infty} \Lambda_{\mathfrak{F}}(d, 2) > 0$ , then*

$$\boxed{\beta_{\mathfrak{F}}^{\text{M}} = \beta_{\mathfrak{F}}^{\text{B}} = \frac{1 + \alpha_{\text{stop}}}{2}}. \quad (26)$$

**Proposition 2.5** (Basic full-class calibration). *For the full bounded convex class and every  $B \geq 2$ ,*

$$cd \leq \Lambda(d, B) \leq Cd^2 \text{polylog}(dB). \quad (27)$$

*The fixed-scale display does not determine the polynomial-scale exponent; the current exponent bracket is recorded in (30) below.*

## 2.2 Adaptive continuation as a genuine separation

For  $C = B_2^d(1)$ ,  $\theta \in S_2^{d-1}(1)$ , and  $s = \sqrt{1 + 2\varepsilon}$ , define

$$F_\theta(x) = \varepsilon + \frac{1}{2}\|x\|^2 - \frac{1}{2}(s - \|\theta - x\|)_+^2. \quad (28)$$

These convex losses are uniquely minimized at  $\theta$  and have minimum zero. We retain this loss family from Bakhtiari et al. (2025) and prove two new facts about it: fresh random directions remain costly even when their radii are adaptive, whereas observation-dependent directions yield a low-value stopped experiment.

**Theorem 2.6** (Adaptive continuation on the dimple family). *At  $\varepsilon = d^{-1}$  and  $b = b_0 \asymp d^{-1}$  in the dimple family (28), take the initial prior  $\Theta \sim \text{Unif}(S_2^{d-1}(1))$  and use the persistent sublevel-set target (8). At this initial state, the one-query restriction and the class that may adapt radii but draws a fresh uniform direction at every query satisfy*

$$\Pi_b^{[1]} = \Omega_\gamma(d^2), \quad \Pi_b^{\text{fresh}} = \Omega_\gamma(d^2), \quad \Pi_b = \tilde{O}_\gamma(d). \quad (29)$$

*Thus observation-dependent direction selection reduces this local complexity by a factor  $\tilde{\Omega}(d)$ . Matching upper bounds for the two restricted classes are not claimed.*

The stopped experiment has three stages. Here  $a$  is the target terminal loss and  $q$  is the current angular error:

| stage                 | progress                                                                                             | (queries, regret)                               |
|-----------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| orthogonal scan       | find $v_0$ with $\langle \theta, v_0 \rangle = \tilde{\Omega}(\max\{\sqrt{\varepsilon}, d^{-1/2}\})$ | $\tilde{O}(d^2/\varepsilon^2, d^2/\varepsilon)$ |
| tangent amplification | use observation-adapted tangent differences to enter a fixed cone                                    | $\tilde{O}(d^2/\varepsilon^2, d^2/\varepsilon)$ |
| angular contraction   | invert boundary observations and repeat $q \mapsto q/2$ until $q = O(a)$                             | $\tilde{O}(d^2/a^2, d^2/a)$                     |

Only the last two stages choose later directions from earlier observations; this is precisely the adaptivity excluded by the fresh-direction lower bound.

## 3 Interpretation and the remaining question

**Consequence of current regret bounds.** For the full bounded convex class, the best currently known regret bounds, combined with Theorem 2.4, give

$$\boxed{\frac{5}{4} \leq \beta_{\mathfrak{F}} \leq \frac{3}{2}}, \quad \boxed{\frac{3}{2} \leq \alpha_{\text{stop}} \leq 2}. \quad (30)$$

The lower endpoint uses the independent theorem of Rajaraman (2026). Its hard support is 1-Lipschitz and anchored at  $f(0) = 0$  on the Euclidean unit ball, so  $g = (f + 1)/2$  is  $[0, 1]$ -valued.



The feedback models are related without decreasing the noise variance: from  $Y = f(x) + \xi$ , draw an independent  $\zeta \sim N(0, 3/4)$  and return  $(Y + 1)/2 + \zeta = g(x) + \xi'$  with  $\xi' \sim N(0, 1)$ . Thus their lower bound embeds in the full bounded class up to universal constants; the known stochastic upper bound supplies the other endpoint. Equation (30) records only the resulting bracket. Neither endpoint is asserted to be sharp, and no part of the present paper reproduces the hidden-curvature construction.

**What the paper’s examples establish.** The following calibrations are proved here:

| family                      | stopped conclusion                                                                                 | interpretation                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| linear losses               | order- $d$ profile                                                                                 | recovers the linear-bandit dimension                                      |
| well-conditioned quadratics | order- $d$ profile                                                                                 | unknown center and orientation do not create a superlinear profile        |
| dimple family               | one-query and fresh-direction values $\Omega(d^2)$ , but unrestricted stopped value $\tilde{O}(d)$ | observation-dependent directions remove a polynomial one-step obstruction |

The first two rows show that unknown orientation alone does not create the full-class gap. The third shows that weak directional information can become easy after it is used to orient later probes. None of these statements determines the full-class exponent.

**Open question 3.1** (Full-class stopped exponent). *Determine  $\alpha_{\text{stop}}$  for the full bounded convex class. Equivalently, by Theorem 2.4, determine its minimax dimension exponent  $\beta_{\mathfrak{F}}$ . Any improvement to either side of (30) must come from a corresponding upper or lower control of the unrestricted stopped profile.*

The profile is invariant under invertible affine changes of variables: such a map transports the target and every stopped experiment while preserving observations, regret, information, and the first-hit event. For every fixed admissible triple  $(c_A, \gamma, \vartheta)$ , the exact characterization also identifies the same stopped exponent  $2\beta_{\mathfrak{F}} - 1$ ; thus the exponent, though not necessarily each finite profile value, is invariant under these threshold constants. These facts support a geometric interpretation. We nevertheless avoid calling the quantity an intrinsic dimension because invariance under alternative target kernels has not been established.

## 4 Scope beyond BCO

Stopped information complexity is not conceptually tied to Gaussian BCO. Its reusable ingredients are a persistent latent environment, nonnegative excess loss, a finite-information target retained under posterior updating, and an online-to-batch terminal recommendation. Candidate extensions include fixed-environment stochastic bandits, noisy optimization, and fixed-law partial monitoring, but each requires its own target, hitting, amortization, and measurability lemmas.

At one state, fractional programming converts the ratio into a Bellman optimal-stopping problem for “cost minus multiplier times progress.” The global upper proof does not solve the full  $T$ -round Bellman program; it glues local solutions through persistent-target information, return-time control, and risk-scale balancing. Convex averaging, Gaussian selection, and the cost  $b^2\mathbb{E}N + b\mathbb{E}R$  are BCO-specific. Thus the proof template is portable, while the exact theorem here is for stationary Gaussian BCO.

## A Proofs for the two-sided stopped-information principle

### A.1 An affine-covariant near-optimal target

Assume that every loss in the prior has range at most  $0 < \Delta \leq 1$  on  $C$ . The analysis needs a near-optimal random index whose information can telescope without depending on an arbitrary choice among multiple minimizers. For  $0 < \eta < \Delta$ , put

$$S_\eta(f) = \{x \in C : f(x) - \min_C f \leq \eta\}, \quad A_\eta^\circ \mid (F = f) \sim \text{Unif}(S_\eta(f)). \quad (31)$$

The conditional draw uses private randomness unavailable to the policy. Near-optimal random learning targets are standard in rate-distortion and satisficing analyses (Dong and Van Roy, 2018; Arumugam and Van Roy, 2021; Russo and Van Roy, 2022). The role here is more specific: the target is affine covariant, retained across repeated local experiments, and paired with a posterior-risk hitting event.

**Lemma A.1** (Sublevel-set near-optimal target). *Equip the continuous losses on  $C$  with the uniform topology. The map  $(f, \eta) \mapsto \text{Unif}(S_\eta(f))$  is a Borel probability kernel on the loss space times  $(0, \Delta)$ . In particular, for every fixed  $\eta$ , the coupling (31) satisfies*

$$F(A_\eta^\circ) - F_C^* \leq \eta \quad \text{almost surely}, \quad I(F; A_\eta^\circ) \leq d \log \frac{\Delta}{\eta}. \quad (32)$$

*If  $T$  is an invertible affine map, then the sublevel-set target for  $f \circ T^{-1}$  on  $T(C)$  has the law of  $T(A_\eta^\circ)$ . Thus the channel is affine covariant and independent of a minimizer representation or of the prior on  $F$ .*

*Proof.* The graph of the correspondence  $(f, \eta) \mapsto S_\eta(f)$  is Borel because  $f \mapsto \min_C f$  is continuous in the uniform norm and  $(f, x) \mapsto f(x)$  is continuous. More explicitly,  $(f, \eta, x) \mapsto \mathbf{1}\{x \in S_\eta(f)\}$  is a bounded jointly Borel function. Its integral against Lebesgue measure is therefore Borel in  $(f, \eta)$ , and the same parameter-integration argument, now treating  $(f, \eta)$  jointly as the parameter and multiplying by the indicator of an arbitrary Borel set in  $C$ , proves that  $(f, \eta) \mapsto \text{Unif}(S_\eta(f))$  is a Borel probability kernel once positivity of the denominator is shown.

Fix any  $y \in \arg \min_C f$  and put  $\alpha = \eta/\Delta$ . For every  $x \in C$ , convexity gives

$$f((1 - \alpha)y + \alpha x) \leq (1 - \alpha)f(y) + \alpha f(x) \leq f_C^* + \eta.$$

Consequently,  $(1 - \alpha)y + \alpha C \subseteq S_\eta(f)$  and  $\text{vol}(S_\eta(f)) \geq \alpha^d \text{vol}(C) > 0$ . The support definition gives the near-optimality statement. Using the uniform law on  $C$  as a reference,

$$I(F; A_\eta^\circ) \leq \mathbb{E} D_{\text{KL}}(P_{A_\eta^\circ \mid F} \parallel \text{Unif}(C)) = \mathbb{E} \log \frac{\text{vol}(C)}{\text{vol}(S_\eta(F))} \leq d \log(1/\alpha).$$

Finally, invertible affine maps carry sublevel sets to sublevel sets and uniform measures to uniform measures, proving covariance.  $\square$

For the reduction, fix a tolerance  $\eta$  and write  $A = A_\eta^\circ$ . On an auxiliary probability space, conditionally on  $F$ , draw the countable family of bucket targets  $A_j \sim A_{c_A b_j}^\circ$ , for a fixed  $c_A \in (0, 1)$ , before the interaction begins, using independent private coins. The learner never observes these coins. Target  $A_j$  is activated for the analysis when bucket  $j$  is first entered and is retained across every later return to that bucket. This ex-ante coupling lets its information drift telescope through the random bucket-entry history.

Let  $h$  be a reached history and write

$$r_B(h) = \inf_{x \in C} \mathbb{E}[F(x) - F_C^* \mid h], \quad \Phi(h) = I(F; A \mid h). \quad (33)$$

Throughout, conditional information at a random history is interpreted statewise:  $I(A; Z \mid H)$  denotes the  $H$ -measurable function which, at  $H = h$ , equals  $D_{\text{KL}}(P_{A,Z|h} \| P_{A|h} \otimes P_{Z|h})$ . An admissible experiment has a complete, almost surely finite stopped transcript  $Z$ , including its length, actions, observations, and private coins. Its actions cannot use the hidden auxiliary randomization in  $A$ . This operational restriction ensures the Markov relation

$$A \perp Z \mid (F, h). \quad (34)$$

In particular, after every history generated without access to the target's private coin, the conditional law of  $A$  given  $F$  remains its original kernel  $K_b(\cdot \mid F)$ .

**Lemma A.2** (Exact information drift). *Every admissible experiment satisfies*

$$\Phi(h) - \mathbb{E}[\Phi(h, Z) \mid h] = I(A; Z \mid h). \quad (35)$$

Consequently, along any adaptive sequence of admissible blocks,

$$\mathbb{E} \sum_k I(A; Z_k \mid H_k) \leq I(F; A). \quad (36)$$

*Proof.* Expand  $I(A; F, Z \mid h)$  in two orders and use  $I(A; Z \mid F, h) = 0$ :

$$I(A; F \mid h) = I(A; Z \mid h) + \mathbb{E}[I(A; F \mid h, Z) \mid h].$$

Iterating this identity proves (36); the remaining conditional mutual information is nonnegative.  $\square$

The information term  $I(A; Z \mid h)$  is the posterior average of the coordinate log gain underlying AIR (Xu and Zeevi, 2025). Under the Markov relation (34), the potential  $I(F; A \mid h)$  has exactly this fixed-index drift, as used by Xu (2026a). The BCO-specific step is to couple the drift to a near-optimal target and the first hitting time of a lower posterior-risk level.

A frequentist high-probability recommendation is not automatically a posterior guarantee: the bound must hold under the terminal posterior on every accepted transcript. The following elementary acceptance rule enforces this condition.

**Lemma A.3** (Posterior-risk acceptance). *Suppose a stopped routine returns  $\hat{x} = \hat{x}(Z)$  and*

$$\mathbb{E}[F(\hat{x}) - F_C^* \mid h] \leq ab.$$

*For  $\gamma > a$ , accept the recommendation only on*

$$\mathbf{M} = \{\mathbb{E}[F(\hat{x}) - F_C^* \mid h, Z] \leq \gamma b\}.$$

*Then every accepted transcript has terminal posterior regret at most  $\gamma b$ , and*

$$\mathbb{P}(\mathbf{M} \mid h) \geq 1 - a/\gamma.$$

*Proof.* The terminal posterior regret is nonnegative and has expectation at most  $ab$  by the tower property. Apply Markov's inequality.  $\square$

The lemma validates accepted transcripts, but does not pay for rejected ones. A complete experiment must still charge its entire expected query and regret cost.

Whenever a capped routine below is analyzed through this rule, we couple it to its full scheduled run but execute its first-hit truncation: after each scalar observation, stop as soon as  $r_B \leq \gamma b$  and return a posterior Bayes action. The gated terminal event implies that a hit occurred by the cap, while truncation only decreases nonnegative query and regret costs. Thus all explicit linear and dimple bounds below concern the first-hit complexity in Definition A.4.

## A.2 Stopped information complexity

Fix constants

$$0 < \gamma < \vartheta < 1 \quad (37)$$

and call  $(h, b)$  *active* when

$$\vartheta b < r_B(h) \leq b. \quad (38)$$

For an experiment  $Z$ , let  $N_Z$  be its length and

$$R_Z = \sum_{t=1}^{N_Z} \{F(X_t) - F_C^*\}.$$

The experiment stops when its goal has already been reached. Starting from  $H_0 = h$ , let

$$\tau_b = \inf\{t \geq 1 : r_B(H_t) \leq \gamma b\}. \quad (39)$$

A *first-hit experiment* chooses a positive adapted stopping time  $\sigma$ , stops at  $\tau = \sigma \wedge \tau_b$ , and has a complete transcript  $Z = H_\tau \setminus H_0$ . It declares  $M = \{\tau_b \leq \sigma\}$  and, on  $M$ , returns a measurable posterior Bayes action  $x_Z$ . Thus

$$\mathbb{E}[F(x_Z) - F_C^* \mid h, Z] = r_B(h, Z) \leq \gamma b \quad \text{on } M. \quad (40)$$

On  $M^c$  the experiment stops before the posterior reaches the smaller scale. No pathwise charge is assigned to either event; the inequalities below control the entire experiment in expectation. The first-hit convention prevents an experiment from continuing after solution merely to harvest target information that the regret problem no longer needs.

The information and risk-reduction terms combine into one statewise quantity.

**Definition A.4** (Stopped information complexity). *Fix an active state  $(h, b, A)$  satisfying (38), where  $A$  is the persistent near-optimal target sampled once for this geometric scale. Let  $\mathcal{B}(h, b, A)$  be the collection of admissible positive-length, almost surely finite first-hit experiments  $(Z, M, x_Z)$  described around (39), with  $\mathbb{E}[N_Z \mid h] < \infty$ . Define*

$$\Pi_b(h; A) = \inf_{(Z, M, x_Z) \in \mathcal{B}(h, b, A)} \frac{b^2 \mathbb{E}[N_Z \mid h] + b \mathbb{E}[R_Z \mid h]}{I(A; Z \mid h) + d\mathbb{P}(M \mid h)}. \quad (41)$$

*The ratio is  $+\infty$  when its denominator is zero. The target's private randomization remains inaccessible to the policy, and the complete transcript includes the stopping time, actions, observations, and policy coins. All complexities below use this first-hit convention.*

Definition A.4 is a stopped analogue of the one-round tradeoff measured by localized and constrained variants of the decision–estimation coefficient (Foster et al., 2021, 2023). It replaces one randomized query by a variable-length BCO experiment whose continuation rule and stopping time are optimized jointly. It also retains a near-optimal target across repeated experiments and counts the first hit of a lower Bayes-risk level as progress. The coordinate log identity underlying AIR supplies the telescoping algebra (Xu and Zeevi, 2025); its fixed-index Bellman extension (Xu, 2026a) supplies the continuation recursion.

**Proposition A.5** (One query and adaptive continuation). *Let  $\mathcal{B}_1(h, b, A)$  be the subclass in Definition A.4 with  $N_Z = 1$  almost surely, and let  $\Pi_b^{[1]}(h; A)$  be the corresponding restricted infimum. Then*

$$\Pi_b(h; A) \leq \Pi_b^{[1]}(h; A). \quad (42)$$

For a one-query law  $\nu$  whose posterior remains active almost surely (so  $\mathbf{M} = \emptyset$ ), write

$$i_\nu = I(A; X, O \mid h), \quad \ell_\nu = \mathbb{E}[F(X) - F_C^* \mid h], \quad \Gamma_\nu = \frac{\ell_\nu^2}{i_\nu}. \quad (43)$$

When  $i_\nu > 0$ ,

$$\frac{b^2 + b\ell_\nu}{i_\nu} = \Gamma_\nu \left\{ \left( \frac{b}{\ell_\nu} \right)^2 + \frac{b}{\ell_\nu} \right\}. \quad (44)$$

Hence, at a state satisfying (38), the two ratios are comparable up to constants on query laws with  $\ell_\nu \leq Kb$ , for fixed  $K < \infty$ . They need not be comparable for high-loss probes.

Assume  $I(F; A \mid h) < \infty$ . For  $\lambda > 0$ , define the one-query offset

$$\mathfrak{D}_{b,\lambda}^{[1]}(h; A) = \inf_{(Z, \mathbf{M}, x_Z) \in \mathcal{B}_1(h, b, A)} [b^2 + b\mathbb{E}[R_Z \mid h] - \lambda \{I(A; Z \mid h) + d\mathbb{P}(\mathbf{M} \mid h)\}]. \quad (45)$$

Then

$$\Pi_b^{[1]}(h; A) \leq \lambda \iff \mathfrak{D}_{b,\lambda}^{[1]}(h; A) \leq 0. \quad (46)$$

For finite problems and experiments capped at  $m$  queries, iterating the same offset with its posterior continuation gives the standard information-penalized optimal-stopping recursion. Thus  $\Pi_b$  is the adaptive optimal-stopping extension of the one-query cost-information tradeoff.

*Proof.* The class inclusion gives (42), and (44) is algebra. Under (38), every query law has  $\ell_\nu \geq r_B(h) > \vartheta b$ ; together with  $\ell_\nu \leq Kb$ , this bounds the factor in braces above and below by positive constants depending only on  $K$  and  $\vartheta$ . Finally, write  $c(Z) = b^2 + b\mathbb{E}[R_Z]$  and  $g(Z) = I(A; Z) + dP(\mathbf{M})$ . Data processing gives the uniform bound  $g(Z) \leq I(F; A \mid h) + d < \infty$ . Hence the infimum of  $c(Z)/g(Z)$  is at most  $\lambda$  exactly when the infimum of  $c(Z) - \lambda g(Z)$  is nonpositive, including the boundary case in which the ratio approaches  $\lambda$  without attaining it. The convention for  $g(Z) = 0$  is harmless because  $c(Z) \geq b^2 > 0$ . This proves (46).  $\square$

**One-round coefficients and stopped experiments.** For one randomized BCO query, the information ratio is  $\ell_\nu^2/i_\nu$ . Offset DEC and constrained regret-DEC likewise optimize one randomized decision and its induced observation, balancing immediate loss against statistical separation from a reference model. The associated algorithms re-solve these one-round problems after each update. Here one round means one native BCO query-observation pair; treating a stopped policy as one abstract decision merely moves the continuation problem into the action space. In the present scalar-feedback model,  $\Pi_b$  instead optimizes every query, adaptation, and stopping decision within a variable-length experiment. We claim no general numerical equivalence between  $\Pi_b$ , DEC, and the information ratio. Equation (44) is the exact algebraic relation for each active information-only one-query law; the stopped value is governed by the optimal-stopping recursion above.

### A.3 Upper and lower transfers

**Theorem A.6** (From local experiments to global regret). *Fix  $0 < c_A < \gamma$ . Conditionally on  $F$ , draw independent private targets  $A_j = A_{c_A b_j}^\circ$  for the geometric risk scales  $b_j = \Delta \vartheta^j$  before interaction, and retain  $A_j$  across every return to scale  $j$ . Suppose that, at every active state, a universally measurable rule selects a first-hit experiment whose ratio in (41) is at most  $\Lambda_d$ , uniformly over scales*

and posterior states. Then, for every  $0 < \eta < \Delta/2$ ,

$$\text{BReg}_T \leq C_{\gamma, \vartheta, c_A} \Lambda_d \frac{d\{1 + \log(\Delta/\eta)\}}{\eta} + 2T\eta, \quad (47)$$

$$\mathbb{E}N_{\text{explore}} \leq C_{\gamma, \vartheta, c_A} \Lambda_d \frac{d\{1 + \log(\Delta/\eta)\}}{\eta^2}. \quad (48)$$

Consequently,

$$\text{BReg}_T = \tilde{O}(\sqrt{d\Lambda_d T}). \quad (49)$$

Thus  $\Lambda_d = \tilde{O}(d^{2-\kappa})$  gives  $\tilde{O}(d^{(3-\kappa)/2}\sqrt{T})$ .

*Proof.* Invoke the selected experiment whenever  $r_B > 2\eta$  and play a posterior Bayes action otherwise. The ratio bound implies, at scale  $b_j$ ,

$$\mathbb{E}[R_Z \mid h] \leq \frac{\Lambda_d}{b_j} \{I(A_j; Z \mid h) + d\mathbb{P}(\mathbf{M} \mid h)\},$$

and the analogous query bound with  $b_j^{-2}$ . For each fixed  $j$ , insert all intervening blocks and posterior-Bayes rounds from other scales and apply Lemma A.2 to the resulting complete history with target  $A_j$ . For every finite number of blocks, discarding the nonnegative information gains outside bucket  $j$  gives the corresponding partial-sum bound. Monotone convergence then yields

$$\mathbb{E} \sum_{k: j(k)=j} I(A_j; Z_k \mid H_k) \leq I(F; A_j).$$

Lemma A.1 bounds the right side by  $d \log\{\Delta/(c_A b_j)\}$ . Geometric summation down to scale  $2\eta$  therefore bounds the information terms by the displayed logarithmic multiples of  $d/\eta$  and  $d/\eta^2$ .

It remains to count risk-reduction outcomes. Condition on one such outcome at scale  $b_j$ , and let  $x$  be the returned action. At every later history,

$$q_t = \mathbb{E}[F(x) - F_C^* \mid H_t]$$

is a bounded nonnegative martingale and initially satisfies  $q_0 \leq \gamma b_j$ . Returning to the same bucket at any later stopping time requires  $r_B(H_t) > \vartheta b_j$ , hence  $q_t > \vartheta b_j$ . Optional stopping, equivalently Doob's maximal inequality for this bounded nonnegative martingale, gives conditional return probability at most  $\gamma/\vartheta$ . Thus the expected number of risk-reduction outcomes in one bucket is at most  $(1 - \gamma/\vartheta)^{-1}$ . Summing  $b_j^{-1}$  and  $b_j^{-2}$  over the geometric buckets gives  $O(\eta^{-1})$  and  $O(\eta^{-2})$ , respectively. Below scale  $2\eta$ , a posterior Bayes action incurs conditional expected regret at most  $2\eta$  per round. If the horizon cuts an experiment short, couple the actual policy to a completed copy used only in the accounting; the executed prefix has no larger nonnegative query or regret cost. Combining these bounds proves (47)–(48); optimizing in  $\eta$  proves (49).  $\square$

The same ratio turns a local bound that holds for every stopped experiment into a Bayes-regret lower bound. This converse is why the lower-bound direction of the core question is stronger than disproving one sufficient condition. For an initial law  $Q$ , write  $\Pi_b(Q; A)$  for the stopped complexity at the empty history.

**Proposition A.7** (From stopped complexity to a regret lower bound). *Suppose an initial prior, a target coupling fixed independently of the policy under test, an active scale  $b$ , and  $\lambda > 0$  satisfy*

$$\Pi_b(Q; A) \geq \lambda. \quad (50)$$

Then, for every integer  $n \geq 1$  and every  $0 < a < \gamma$ , every  $n$ -round adaptive policy satisfies

$$\text{BReg}_n \geq \min \left\{ abn, \left[ \frac{\lambda d(1 - a/\gamma)}{b} - bn \right]_+ \right\}. \quad (51)$$

In particular, if  $d\lambda/b^2 \geq 8$ , then at

$$n = \left\lfloor \frac{d\lambda}{4b^2} \right\rfloor \quad (52)$$

every  $n$ -round adaptive policy has

$$\text{BReg}_n \geq c_\gamma \frac{d\lambda}{b} = \Omega_\gamma(\sqrt{d\lambda n}). \quad (53)$$

Because this is a Bayes lower bound for an initial prior, it immediately lower-bounds the minimax regret of every loss class containing  $\text{supp}(Q)$ . At a reached state the same argument gives only a conditional continuation bound unless the state is made initial or is reached with controlled probability.

*Proof.* Fix a policy and let  $R_n = \mathbb{E} \sum_{t=1}^n \{F(X_t) - F_C^*\}$ . If  $R_n > abn$ , the first term in (51) applies. Otherwise put  $\bar{X} = n^{-1} \sum_{t=1}^n X_t$ . Convexity gives

$$\mathbb{E}[F(\bar{X}) - F_C^*] \leq R_n/n \leq ab.$$

Let

$$q_n = \mathbb{E}[F(\bar{X}) - F_C^* \mid H_n].$$

Markov's inequality gives  $\mathbb{P}(q_n \leq \gamma b) \geq 1 - a/\gamma$ . On this event  $r_B(H_n) \leq q_n$ , so the first-hit time  $\tau_b = \inf\{t \geq 1 : r_B(H_t) \leq \gamma b\}$  is at most  $n$ . Couple the policy to the first-hit block with  $\tau = n \wedge \tau_b$  and  $M = \{\tau_b \leq n\}$ . Nonnegativity of regret gives  $\mathbb{E}\tau \leq n$  and  $\mathbb{E}R_\tau \leq R_n$ , while  $\mathbb{P}(M) \geq 1 - a/\gamma$ . Hence (50) gives

$$b^2 n + bR_n \geq b^2 \mathbb{E}\tau + b\mathbb{E}R_\tau \geq \lambda \{I(A; Z_\tau) + d\mathbb{P}(M)\} \geq \lambda d(1 - a/\gamma),$$

which proves (51). Take  $a = \gamma/2$  and then (52); the assumption  $d\lambda/b^2 \geq 8$  absorbs the floor and gives (53).  $\square$

*Proof of Theorem 2.1.* Fix a prior and choose a universally measurable block rule with ratio at most  $1 + \Lambda_{\mathfrak{F}}^{\text{policy}}(d, T)$  on all active scales  $b \geq T^{-1}$ . Theorem A.6, with  $\Delta = 1$ , gives (17) for every  $\eta \geq T^{-1}$ . Lemma C.1 replaces the resulting finite-horizon policy, for this prior, by a Borel policy with the same Bayes law. Taking the supremum over priors proves the displayed worst-prior Bayes bound. Optimizing  $\eta$ , with the trivial regret bound  $T$  in the complementary regime, yields (18).

Conversely, the strict inequality  $\lambda < \Lambda_{\mathfrak{F}}^{\text{state}}(d, B)$  supplies an active state  $(Q, b)$  whose every first-hit experiment has ratio larger than  $\lambda$ . Treat  $Q$  as an initial prior and apply Proposition A.7. This gives (19) and, at the calibrated horizon, (20)–(21).

For the exponent step take  $B = d^q$ . Then  $b \geq d^{-q}$ , while the general profile bound in Proposition A.11, applied to the computation-unrestricted stochastic-BCO regret bound, gives  $\Lambda_{\mathfrak{F}}^{\text{state}}(d, d^q) \leq d^{2+o(1)}$  for every subclass of bounded convex losses. Hence  $\lambda \leq d^{2+o(1)}$ , and the calibrated horizon is at most  $d^{2q+3+o(1)}$ . It lies in a polynomial horizon window; taking dimension growth rates proves (22).  $\square$



#### A.4 Linear and quadratic calibrations

The next proposition calibrates the smallest possible universal stopped complexity. It controls arbitrary adaptive measurements, including information about joint parities or other nonlinear functions of the hidden signs.

**Proposition A.8** (A linear lower bound). *Let  $C = B_2^d(1)$ , let  $0 < b \leq 1/2$ , draw  $V \sim \text{Unif}\{-1, 1\}^d$ , put  $\theta_V = V/\sqrt{d}$ , and define*

$$F_V(x) = b\{1 - \langle \theta_V, x \rangle\}. \quad (54)$$

*Write  $Q_{\text{lin}}$  for this prior. Then  $F_V \in [0, 2b]$ ,  $F_V^* = 0$ , and the initial Bayes risk is  $b$ . For every randomized adaptive stopped transcript  $Z$  with random length  $N_Z$  satisfying  $\mathbb{E}N_Z < \infty$ , and every target  $A$  obtained from  $(V, U)$  with private randomization  $U$  conditionally independent of  $Z$  given  $V$ ,*

$$I(A; Z) \leq I(V; Z) \leq \frac{b^2}{d} \mathbb{E}N_Z. \quad (55)$$

*If  $(Z, \mathbf{M}, x_Z)$  satisfies the posterior risk bound at level  $\gamma b$ , then*

$$\mathbb{E}N_Z \geq c_\gamma \left\{ \frac{d}{b^2} I(A; Z) + \frac{d^2}{b^2} \mathbb{P}(\mathbf{M}) \right\}. \quad (56)$$

*Consequently*

$$\Pi_b(Q_{\text{lin}}; A) \geq c_\gamma d. \quad (57)$$

*Consequently every universal stopped-complexity upper bound must be at least of order  $d$ .*

*Proof.* Write  $P_v$  for the law of the complete stopped transcript under  $V = v$ , let  $\mu$  be uniform on the cube, and let  $\bar{P} = 2^{-d} \sum_v P_v$ . The tensorized two-point entropy inequality states that, for every positive  $f$  on the cube,

$$\text{Ent}_\mu(f) \leq \frac{1}{4} \sum_{i=1}^d \mathbb{E}_\mu[(f(V) - f(V^{(i)}))(\log f(V) - \log f(V^{(i)}))],$$

where  $V^{(i)}$  flips coordinate  $i$ . Apply it pointwise to  $f_z(v) = dP_v/d\bar{P}(z)$  after replacing  $f_z$  by  $f_z + \varepsilon$ , and let  $\varepsilon \downarrow 0$ ; this handles densities that vanish. Integrating in  $z$  gives

$$I(V; Z) \leq \frac{1}{4} 2^{-d} \sum_{v,i} \{D_{\text{KL}}(P_v \| P_{v^{(i)}}) + D_{\text{KL}}(P_{v^{(i)}} \| P_v)\}. \quad (58)$$

For adaptive Gaussian observations, apply the finite-horizon chain rule to  $N_Z \wedge m$  and let  $m \rightarrow \infty$  by monotone convergence. This yields

$$D_{\text{KL}}(P_v \| P_{v^{(i)}}) = \frac{2b^2}{d} \mathbb{E}_v \sum_{t \leq N_Z} X_{t,i}^2.$$

Average this identity in (58) and use  $\|X_t\|_2 \leq 1$  to obtain the second inequality in (55); the first is data processing.

On a terminal transcript  $z \in \mathbf{M}$ , let  $m_z = \mathbb{E}[\theta_V \mid Z = z]$ . The posterior risk bound gives

$$1 - \langle m_z, x_Z \rangle \leq \gamma, \quad \|m_z\|_2 \geq 1 - \gamma.$$



The law of  $\theta_V$  is  $1/d$ -sub-Gaussian in every unit direction. The variational formula for relative entropy therefore gives

$$D_{\text{KL}}(P_{V|Z=z}||\mu) \geq \frac{d}{2}\|m_z\|_2^2.$$

Averaging over  $z \in \mathbf{M}$  shows

$$I(V; Z) \geq \frac{d}{2}(1 - \gamma)^2 \mathbb{P}(\mathbf{M}). \quad (59)$$

Combine (55) and (59). They respectively bound  $dI(A; Z)$  and  $d^2 \mathbb{P}(\mathbf{M})$  by constant multiples of  $b^2 \mathbb{E} N_Z$ , proving (56) and then (57).  $\square$

The order- $d$  value is attained throughout the unit-ball linear class, not only on the preceding prior.

**Proposition A.9** (Stopped complexity of linear losses). *Fix  $0 < \beta \leq 1/2$  and consider*

$$F_\theta(x) = \beta\{1 - \langle \theta, x \rangle\}, \quad x \in B_2^d(1), \quad \theta \in S_2^{d-1}(1). \quad (60)$$

*For every posterior  $Q$  on  $\Theta$ , every active risk scale  $a$  with  $\vartheta a < r_B(Q) \leq a$ , and every admissible target coupling  $A$ , one has*

$$\Pi_a(Q; A) \leq C_{\gamma, \vartheta} d. \quad (61)$$

*Together with Proposition A.8, this shows that the worst stopped complexity within the class (60) is  $\Theta_{\gamma, \vartheta}(d)$  up to universal constants. Thus the full-class superlinear obstruction implied by (30) must use structure absent from this fixed-amplitude Euclidean class.*

*Proof.* Put  $u = \min\{a, \beta\}$  and

$$m = \left\lceil C_0 \frac{d}{\beta u} \right\rceil,$$

where  $C_0 = C_0(\gamma)$  is chosen below. Query each coordinate action  $e_i$  exactly  $m$  times. If  $\bar{O}_i$  is its averaged observation, set

$$\tilde{\theta}_i = 1 - \bar{O}_i/\beta, \quad \hat{x} = \frac{\tilde{\theta}}{\|\tilde{\theta}\|_2},$$

with an arbitrary definition on the null event  $\tilde{\theta} = 0$ . Conditionally on every  $\theta$ ,

$$\mathbb{E}_\theta \|\tilde{\theta} - \theta\|_2^2 = \frac{d}{m\beta^2}.$$

For every nonzero vector  $y$  and unit vector  $\theta$ ,

$$\left\| \frac{y}{\|y\|_2} - \theta \right\|_2 \leq 2\|y - \theta\|_2,$$

because  $|\|y\|_2 - 1| \leq \|y - \theta\|_2$ . Therefore, uniformly in  $\theta$ ,

$$\mathbb{E}_\theta F_\theta(\hat{x}) = \frac{\beta}{2} \mathbb{E}_\theta \|\hat{x} - \theta\|_2^2 \leq \frac{2d}{m\beta} \leq \frac{2u}{C_0}.$$

Choose  $C_0 \geq 8/\gamma$ . Under every posterior the returned action has expected regret at most  $\gamma a/4$ . Posterior-risk acceptance at scale  $a$  therefore gives an event  $\mathbf{G}$  of probability at least  $3/4$  on which  $\hat{x}$  has posterior risk at most  $\gamma a$ . Under the first-hit convention,  $\mathbf{G} \subseteq \mathbf{M}$ , so  $\mathbb{P}(\mathbf{M}) \geq 3/4$  and actual prefix costs are no larger than the caps below.

The block has deterministic length  $N = dm$ , and every query has regret at most  $2\beta$ , so  $R_Z \leq 2\beta N$ . If  $a \leq \beta$ , then

$$a^2 N + a \mathbb{E} R_Z \leq C_1 d^2.$$

If  $a > \beta$ , the origin has regret exactly  $\beta$ , so activity implies  $\beta \geq r_B(Q) > \vartheta a$ . The same display then holds with  $C_1$  enlarged by a factor depending only on  $\vartheta$ . Since the denominator of (41) is at least  $d\mathbb{P}(\mathbf{M}) \geq 3d/4$ , the block proves (61). The lower bound follows by taking  $Q$  to be the Rademacher prior of Proposition A.8 and  $a = \beta$ .  $\square$

The linear calculation is not an artifact of a flat loss surface. The same exponent persists when both the minimizer and the orientation of a curved loss are unknown. This is a calculation of the stopped profile, not the first favorable-regime  $d\sqrt{T}$  result for curved losses: Ito (2020) obtained optimal dimension dependence for smooth, strongly convex online BCO under an interior-minimizer condition.

**Proposition A.10** (Uniformly well-conditioned quadratics have exponent one). *Fix  $R \geq 1$  and, on  $C = B_2^d(1)$ , consider*

$$\mathfrak{Q}_R = \left\{ F_{H,y}(x) = \frac{1}{4R}(x-y)^\top H(x-y) : I \preceq H \preceq RI, \|y\|_2 \leq \frac{1}{4} \right\}. \quad (62)$$

*Every member of  $\mathfrak{Q}_R$  takes values in  $[0, 25/64]$ . For every posterior on  $\mathfrak{Q}_R$ , every active scale  $a$ , and every admissible private target coupling  $A$ ,*

$$\Pi_a(Q; A) \leq C_{R,\gamma,\vartheta} d. \quad (63)$$

*At a constant active scale, a shifted-hypercube subfamily satisfies, for every such target,*

$$\Pi_a(Q; A) \geq c_{R,\gamma} d.$$

*Consequently, for every  $B \geq 64R$ , both profiles of  $\mathfrak{Q}_R$  are  $\Theta_R(d)$ . In particular, both stopped-complexity exponents equal one.*

*Proof.* Write  $F(x) = \frac{1}{2}(x-y)^\top B(x-y)$ , where  $\mu I \preceq B \preceq LI$  with  $\mu = (2R)^{-1}$  and  $L = 1/2$ . Project all iterates onto  $B_2^d(1/2)$ , start from zero, and put  $h = 1/4$ . At iteration  $k$ , query  $x_k \pm he_i$  once for every coordinate and use the central differences as a gradient estimate  $\hat{g}_k$ . All query points lie in  $C$ , and quadraticity gives

$$\mathbb{E}[\hat{g}_k \mid x_k, F] = \nabla F(x_k), \quad \mathbb{E}[\|\hat{g}_k - \nabla F(x_k)\|_2^2 \mid x_k, F] = 8d.$$

Projected stochastic gradient descent with  $\eta_k = 2/\{\mu(k+k_0)\}$  and sufficiently large  $k_0 = k_0(R)$  obeys the standard strongly-convex recursion

$$\mathbb{E}\|x_k - y\|_2^2 \leq C_R \min \left\{ 1, \frac{d}{k+k_0} \right\}.$$

Indeed, nonexpansiveness of projection, strong convexity, and  $\|\nabla F(x)\| \leq L\|x-y\|$  give this bound by induction from

$$\mathbb{E}[\|x_{k+1} - y\|^2 \mid x_k, F] \leq (1 - 2\mu\eta_k + L^2\eta_k^2)\|x_k - y\|^2 + 8d\eta_k^2.$$

Choose  $k_0$  so that  $L^2\eta_k^2 \leq \mu\eta_k/2$ . Writing  $s_k = \mathbb{E}\|x_k - y\|^2$ , the preceding display becomes

$$s_{k+1} \leq \left( 1 - \frac{3}{k+k_0} \right) s_k + \frac{C_R d}{(k+k_0)^2}.$$

The claim follows by induction with  $s_k \leq C_R d/(k + k_0)$  after enlarging  $C_R$ ; the trivial bound  $s_k \leq (3/2)^2$  supplies the initial range. Taking  $K = \lceil C_{R,\gamma} d/a \rceil$  makes  $\mathbb{E}_F F(x_K) \leq \gamma a/16$ , uniformly in  $F$ . Moreover the exact pair identity

$$\sum_{i=1}^d \{F(x_k + h e_i) + F(x_k - h e_i)\} = 2dF(x_k) + h^2 \operatorname{tr}(B)$$

and the preceding decay imply

$$N_Z = 2dK \leq C_{R,\gamma} \frac{d^2}{a}, \quad \mathbb{E}_F R_Z \leq C_{R,\gamma} \frac{d^2}{a}.$$

Indeed  $\mathbb{E} F(x_k) \leq C_R \min\{1, d/(k + k_0)\}$  and

$$\sum_{k=1}^K \min\left\{1, \frac{d}{k + k_0}\right\} \leq C_R d \{1 + \log(1/a)\}.$$

Thus the  $2dF(x_k)$  terms contribute at most  $O_R(d^2 \{1 + \log(1/a)\}) = O_R(d^2/a)$ . The common curvature term  $h^2 \operatorname{tr}(B) = O_R(d)$  contributes  $O_R(dK) = O_R(d^2/a)$ . The bounds hold under every posterior. Posterior-risk acceptance and first-hit truncation therefore give  $\mathbb{P}(\mathbf{M}) \geq 15/16$ . Using only the hit term in the denominator of (41) proves (63).

For the converse, fix  $H = I$ , draw  $V \sim \operatorname{Unif}\{-1, 1\}^d$ , and set  $y_V = rV/\sqrt{d}$  with  $r = 1/4$ . The initial Bayes action is zero and its risk is the constant  $b = r^2/(4R)$ . Flipping bit  $i$  changes a query mean by  $-rv_i x_i/(R\sqrt{d})$ . The cube-entropy comparison and stopped Gaussian chain rule used in Proposition A.8 therefore give

$$I(V; Z) \leq \frac{C_R}{d} \mathbb{E} N_Z.$$

If  $m_Z = \mathbb{E}[V \mid Z]$ , the terminal posterior Bayes risk is  $b\{1 - \|m_Z\|^2/d\}$ . Hence on  $\mathbf{M}$  one has  $\|m_Z\|^2 \geq (1 - \gamma)d$ , and the sub-Gaussian entropy variational bound yields  $I(V; Z) \geq c_\gamma d \mathbb{P}(\mathbf{M})$ . Finally  $I(A; Z) \leq I(V; Z)$  by data processing. Thus

$$I(A; Z) + d \mathbb{P}(\mathbf{M}) \leq \frac{C_{R,\gamma}}{d} \mathbb{E} N_Z,$$

whereas the numerator in (41) is at least  $b^2 \mathbb{E} N_Z$ . This proves the order- $d$  lower bound. The explicit upper routine and this constant-scale prior give the two matching statements.  $\square$

## A.5 From regret bounds to stopped experiments

**Proposition A.11** (A stopped experiment from a regret bound). *Suppose an algorithm has, uniformly over fixed losses and all  $n, \delta$ , a high-probability regret bound*

$$R_n \leq \{D\sqrt{n} + A\} L(n, d, \delta)$$

*outside an event of probability  $\delta$ , where  $A \geq 0$  and, for fixed constants  $C_L, q < \infty$ ,*

$$L(n, d, \delta) \leq C_L [\log\{\operatorname{end}/\delta\}]^q. \tag{64}$$

*Then, from any posterior state and scale  $b$ , it yields a first-hit experiment with*

$$\mathbb{E} N_Z \leq \tilde{O}(1 + D^2/b^2 + A/b) \mathbb{P}(\mathbf{M}), \quad \mathbb{E} R_Z \leq \tilde{O}(b + D^2/b + A) \mathbb{P}(\mathbf{M}), \quad \mathbb{P}(\mathbf{M}) \geq \frac{1}{2}.$$

In particular, the computation-unrestricted bound of [Fokkema et al. \(2024, Theorem 2\)](#) for arbitrary bodies, and their efficient bound for symmetric bodies, have  $D = Md^2 = \tilde{O}(d^{3/2})$  and  $A = 0$ , since in these two cases  $M = \tilde{O}(d^{-1/2})$ . Substitution above gives stopped complexity  $\tilde{O}(D^2/d) = \tilde{O}(d^2)$ . Their efficient arbitrary-body choice  $M = \tilde{O}(d^{-1/4})$  instead has  $D = \tilde{O}(d^{7/4})$ .

*Proof.* Take  $\delta = \gamma b/4$  and let  $n$  be the least integer for which  $DL(n, d, \delta)/\sqrt{n} + AL(n, d, \delta)/n \leq \gamma b/4$ . The explicit bound (64) and a standard doubling argument give

$$n \leq C_{\gamma, C_{L,q}} \left( \frac{D^2}{b^2} + \frac{A}{b} + 1 \right) \left[ \log \left\{ ed \left( \frac{D^2}{b^3} + \frac{A}{b^2} + \frac{1}{b} \right) \right\} \right]^{2q+2}.$$

Thus  $n = \tilde{O}(1 + D^2/b^2 + A/b)$ . Since regret is at most  $n$  on the failure event,  $\mathbb{E}R_n \leq \gamma b n/2$ . For  $\bar{X} = n^{-1} \sum_{t=1}^n X_t$ , convexity gives

$$\mathbb{E}[F(\bar{X}) - F_C^*] \leq \mathbb{E}R_n/n \leq \gamma b/2.$$

Apply Lemma A.3 with  $a = \gamma/2$  to the terminal posterior regret of  $\bar{X}$ . Then  $\mathbb{P}(\mathbf{M}) \geq 1/2$ , so the deterministic schedule can be truncated at the first posterior state with Bayes risk at most  $\gamma b$ . The gated terminal event implies that such a hit has occurred, while prefix length and regret can only decrease. Since  $\mathbb{P}(\mathbf{M}) \geq 1/2$ , the schedule cap and expected regret are absorbed by the displayed hit-probability terms.  $\square$

## B A multistep separation on the dimple family

The dimple construction of [Bakhtiari et al. \(2025, Lemma 10\)](#) makes the distinction between adaptive radii and observation-dependent direction selection exact. The following consequence is self-contained except for the known convexity of the displayed losses. It rules out a natural class of local searches; it is not a lower bound for unrestricted stochastic BCO.

**Proposition B.1** (Lower bound for nonadaptive designs and fresh uniform directions). *Let  $d$  be sufficiently large,  $C = B_2^d(1)$ ,  $d^{-1} \leq \varepsilon \leq \varepsilon_0$  for a small universal constant  $\varepsilon_0$ ,  $s = \sqrt{1 + 2\varepsilon}$ , and  $\Theta \sim \text{Unif}(S_2^{d-1}(1))$ . Recall the family  $F_\theta$  from (28); its losses take values in  $[0, 1]$ . Their prior Bayes action is  $z = 0$  and*

$$b_0 = F_\theta(0) = s - 1 = \frac{2\varepsilon}{\sqrt{1 + 2\varepsilon} + 1} \asymp \varepsilon, \quad \nabla F_\theta(0) = -b_0\theta. \quad (65)$$

*Every nonadaptive batch  $X_1, \dots, X_N$ , possibly randomized independently of  $\Theta$  but fixed before its observations, satisfies*

$$I(\Theta; O_{1:N} \mid X_{1:N}) \leq CN(\varepsilon^4 + d^{-4}). \quad (66)$$

*The same bound holds sequentially for any adaptive experiment which, at time  $t$ , first draws  $B_t \sim \text{Unif}(S_2^{d-1}(1))$  independently of the past and of  $\Theta$ , then chooses an arbitrary history- and  $B_t$ -measurable radius  $R_t \in [0, 1]$ , queries  $X_t = R_t B_t$ , and observes  $O_t = F_\Theta(X_t) + \xi_t$ . For every current posterior,*

$$I(\Theta; B_t, O_t \mid H_t) \leq C(\varepsilon^4 + d^{-4}). \quad (67)$$

*Consequently, for this fresh-uniform-direction class, a stopped transcript  $Z$  of expected length  $N$  satisfies  $I(\Theta; Z) \leq CN(\varepsilon^4 + d^{-4})$ .*

*If an event  $\mathbf{M} \in \sigma(Z)$  reports an action  $x_Z$  with posterior regret at most  $\gamma b_0$ , where  $\gamma < 1$ , then*

$$I(\Theta; Z) \geq \frac{d-1}{2}(1-\gamma)^2 \mathbb{P}(\mathbf{M}). \quad (68)$$

Thus either  $\Omega(d)$  information or a constant-probability target-risk hit requires

$$N = \Omega_\gamma \left( \frac{d}{\varepsilon^4 + d^{-4}} \right) = \Omega_\gamma(d/\varepsilon^4), \quad (69)$$

whereas the sample scale corresponding to complexity  $O(d)$  at  $b_0 \asymp \varepsilon$  is  $d^2/\varepsilon^2$ . The restricted class loses the polynomial factor  $1/(d\varepsilon^2)$  whenever  $d^{-1} \lesssim \varepsilon \ll d^{-1/2}$ . Its direction law, not merely its radii, must respond to the observations.

*Proof.* Put  $q(x) = \varepsilon + \|x\|^2/2$ . For  $u = \langle \theta, B \rangle$ ,

$$|F_\theta(rB) - q(rB)| = \frac{1}{2} (s - \sqrt{1 + r^2 - 2ru})_+^2.$$

Minimizing the square root over  $r \in [0, 1]$  shows that the supremum of the right side is  $(s - 1)^2/2$  when  $u \leq 0$ , and  $(s - \sqrt{1 - u^2})^2/2$  when  $u > 0$ . On  $|u| \leq 1/2$ , its square is at most  $C(\varepsilon^4 + u^8)$ ; the complementary spherical cap has exponentially small mass. Since

$$\mathbb{E}u^8 = \frac{105}{d(d+2)(d+4)(d+6)},$$

rotation invariance gives, uniformly in  $\theta$ ,

$$\mathbb{E}_B \sup_{0 \leq r \leq 1} |F_\theta(rB) - q(rB)|^2 \leq C(\varepsilon^4 + d^{-4}). \quad (70)$$

For a nonadaptive batch, condition on its public random seed, apply the same bound after interchanging the Haar variables  $\Theta$  and  $B$  by rotational invariance for each fixed design direction, and then apply the Gaussian KL bound relative to the baseline observations  $q(X_t) + \xi_t$ , and sum (70) over  $t$ . For the sequential class, the same estimate remains true after replacing the prior of  $\Theta$  by an arbitrary posterior because the fresh  $B_t$  is uniform. Conditional Gaussian-channel capacity, with the known  $q(R_t B_t)$  subtracted, proves (67); the stopped chain rule proves the transcript bound.

Convexity at the origin and (65) give

$$F_\theta(x) \geq b_0 - b_0 \langle \theta, x \rangle.$$

On an accepted transcript, with  $m_Z = \mathbb{E}[\Theta \mid Z]$ , posterior regret at most  $\gamma b_0$  therefore forces  $\|m_Z\| \geq 1 - \gamma$ . The uniform law  $\sigma$  on the sphere is  $1/(d-1)$ -sub-Gaussian, so the variational formula gives

$$D_{\text{KL}}(Q \parallel \sigma) \geq \frac{d-1}{2} \|\mathbb{E}_Q \Theta\|^2.$$

Apply this to each posterior and average over  $\mathbf{M}$  to obtain (68).  $\square$

The next result shows that this restricted obstruction is removed when the probe directions may depend on the observations.

**Theorem B.2** (Observation-adaptive experiment for the dimple family). *For the family (28), every  $d^{-1} \leq \varepsilon \leq \varepsilon_0$ , and all sufficiently large  $d$ , there are universal constants  $0 < \gamma_0 < \gamma < 1$  such that, for every target scale  $0 < a \leq b_0$ , a capped observation-adaptive experiment satisfies, uniformly over  $\theta \in S_2^{d-1}(1)$ ,*

$$\mathbb{P}_\theta \{F_\theta(\hat{x}) \leq \gamma_0 a\} \geq 1 - ca, \quad N_Z \leq \tilde{O}(d^2/a^2), \quad \mathbb{E}_\theta R_Z \leq \tilde{O}(d^2/a). \quad (71)$$

For every prior on  $\Theta$ , applying posterior-risk acceptance to  $\hat{x}$  therefore produces an event  $\mathbf{G}$  of constant probability on which  $\hat{x}$  has terminal posterior regret at most  $\gamma a$ . The first-hit event  $\mathbf{M}$  contains  $\mathbf{G}$ . For  $b \leq b_0$  run the routine at  $a = b$ ; for  $b_0 < b < b_0/\gamma$  run it at  $a = b_0$ ; and for  $b \geq b_0/\gamma$ , no posterior state is active because its Bayes risk is at most  $b_0 \leq \gamma b$ . These experiments have stopped complexity  $\tilde{O}(d)$  throughout the active range. At the critical scale  $a = b_0 \asymp \varepsilon$ , the adaptive cost is

$$N_Z = \tilde{O}(d^2/\varepsilon^2), \quad \mathbb{E}R_Z = \tilde{O}(d^2/\varepsilon). \quad (72)$$

The restricted lower bound in Proposition B.1 exceeds the  $O(d)$  benchmark by the factor  $1/(d\varepsilon^2)$  whenever that factor diverges, and the adaptive routine matches the target up to polylogarithmic factors. At the Bakhtiari–Lattimore–Szepesvári scale  $\varepsilon = 8 \log(d)/d$ , the respective query bounds are  $\tilde{O}(d^4/\log^2 d)$  and  $\Omega(d^5/\log^4 d)$ , a separation by a factor  $\tilde{\Omega}(d)$ .

At  $\varepsilon = d^{-1}$ , these bounds yield the following separation. Let  $\Pi_b^{\text{fresh}}$  denote Definition A.4 with the infimum restricted to the sequential fresh-uniform-direction policies in Proposition B.1. Every one-query experiment is covered by the  $N = 1$  nonadaptive information bound.

*Proof of Theorem 2.6.* For a one-query experiment, the  $N = 1$  case of (66) gives the first bound below. For a fresh-uniform-direction first-hit experiment, (67) and the stopped chain rule give the second:

$$I(\Theta; Z) \leq Cd^{-4} \quad \text{or} \quad I(\Theta; Z) \leq Cd^{-4}\mathbb{E}N_Z, \quad I(\Theta; Z) \geq c_\gamma d\mathbb{P}(\mathbf{M}).$$

The last inequality gives  $d\mathbb{P}(\mathbf{M}) \leq C_\gamma I(\Theta; Z)$ . The sublevel-set target is a randomized function of  $\Theta$  whose private coin is inaccessible, so  $I(A; Z) \leq I(\Theta; Z)$ . Hence

$$I(A; Z) + d\mathbb{P}(\mathbf{M}) \leq C_\gamma d^{-4}\mathbb{E}N_Z,$$

where  $\mathbb{E}N_Z = 1$  in the one-query case. Since  $b^2 \asymp d^{-2}$ , the numerator of (41) is at least  $cd^{-2}\mathbb{E}N_Z$ , proving the two restricted lower bounds. The risk-reduction experiment in Theorem B.2 has  $b^2\mathbb{E}N_Z + b\mathbb{E}R_Z = \tilde{O}(d^2)$  and  $\mathbb{P}(\mathbf{M}) \geq c_\gamma$ , which proves the unrestricted upper bound.  $\square$

The proof, given in Appendix D, has three stages. A small centered orthogonal design creates correlation  $\tilde{\Omega}(\max\{\sqrt{\varepsilon}, d^{-1/2}\})$ . Tangent finite differences amplify this to a fixed angle. On the unit sphere the active loss has the exact form

$$F_\theta(x) = \psi(\|\theta - x\|), \quad \psi(r) = sr - r^2/2,$$

so observation-adapted boundary probes invert the tangent coordinates and halve the angular error geometrically down to  $O(a)$ . This last loss-sensitive stage is essential: constant correlation alone does not give loss at the target scale.

The gain comes from choosing later query directions using earlier observations; adapting only the radii does not suffice. This is a family-specific result, not a minimax lower or upper bound for unrestricted BCO.

## C Technical completion: measurable implementation and duality

The results in this section close admissibility and value-level quantifiers; they do not enter the scale-amortization or first-hit arguments that drive the dimension dependence.

**Lemma C.1** (Prior-wise Borel realization). *Fix a Borel prior  $Q$  and a finite horizon  $T$ . On standard Borel history, action, and observation spaces, every universally measurable randomized nonanticipating policy has a Borel randomized nonanticipating policy that induces the same joint law of  $(F, H_T)$  under  $Q$ . The conclusion also holds when the action space is augmented by a stopping symbol.*

*Proof.* Represent the time- $t$  action kernel as a universally measurable map from the history space into the standard Borel space of probability measures on the action space. Under the history marginal reached at time  $t - 1$ , such a map has a Borel version: apply the defining completion property to a countable generator of the target standard Borel space and use a Borel coding of that space. Replace the kernel by this version. The replacement agrees almost surely under the reached history law and therefore leaves the joint law through time  $t$  unchanged. Induction over  $t = 1, \dots, T$  gives a Borel policy with the same strategic law. A stopping decision is handled by adjoining one isolated point to the action space. In particular, every finite-horizon Bayes regret under  $Q$  is unchanged.  $\square$

### C.1 Bayes–minimax duality for the Gaussian model

*Proof of Proposition 2.2.* Let  $\gamma = N(0, 1)$  and  $H_T = (C \times \mathbb{R})^T$ . Under the reference observation law  $\gamma$ , let  $\mathcal{S}_T$  be the set of strategic measures

$$P(dh_T) = \prod_{t=1}^T \pi_t(dx_t \mid h_{t-1}) \gamma(dy_t), \quad (73)$$

where the  $\pi_t$  are Borel kernels. This set is convex and weakly compact. Indeed,  $C$  is compact and every  $P \in \mathcal{S}_T$  has  $Y_{1:T}$ -marginal  $\gamma^{\otimes T}$ , so the family is tight. It is closed because, for every bounded continuous  $a$  and  $b$ ,

$$\int a(h_{t-1}, x_t) b(y_t) dP = \left( \int b d\gamma \right) \int a(h_{t-1}, x_t) dP, \quad (74)$$

and these identities pass to weak limits. Conversely, standard-Borel disintegration identifies every  $P \in \mathcal{S}_T$  with a randomized nonanticipating Borel policy. Gaussian likelihoods are strictly positive, so policies identified outside a reference-null set have the same risk under every  $f$ .

For  $f \in \mathfrak{F}_d(C)$  define

$$L_f(h_T) = \exp \left\{ \sum_{t=1}^T (y_t f(x_t) - \tfrac{1}{2} f(x_t)^2) \right\}, \quad (75)$$

$$u(P, f) = \int L_f(h_T) \sum_{t=1}^T \{f(x_t) - \min_C f\} dP(h_T). \quad (76)$$

The Gaussian change-of-measure formula gives  $u(P, f) = \mathbb{E}_f^\pi R_T$ . Moreover,  $u(\cdot, f)$  is affine and continuous on  $\mathcal{S}_T$ , while  $u(P, \cdot)$  is continuous in the uniform topology. The integrand is continuous and bounded by

$$T \exp \left( \sum_{t=1}^T |y_t| \right),$$

which is uniformly integrable over  $\mathcal{S}_T$  because all its members have the same Gaussian  $Y_{1:T}$ -marginal.



Let  $\mathcal{Q}_0$  be the convex set of finitely supported priors on  $\mathfrak{F}_d(C)$  and put  $u(P, Q) = \int u(P, f)Q(df)$ . Sion's theorem (Sion, 1958), applied to the compact convex set  $\mathcal{S}_T$  and the convex set  $\mathcal{Q}_0$ , yields

$$\min_{P \in \mathcal{S}_T} \sup_f u(P, f) = \sup_{Q \in \mathcal{Q}_0} \min_{P \in \mathcal{S}_T} u(P, Q). \quad (77)$$

The left side is minimax regret, and the right side is worst Bayes regret over finitely supported priors. Allowing all Borel priors cannot decrease the latter value and, by weak duality, cannot increase it beyond the left side. This proves (24) and the finite-support assertion.  $\square$

## C.2 Measurable selection of stopped experiments

We use two standard facts about relative entropy. They are recorded here because the selection argument depends on measurability of the information term and on approximation by finite prefixes.

**Lemma C.2** (Relative entropy under measurable limits). *On a Polish space,  $(P, Q) \mapsto D_{\text{KL}}(P\|Q)$  is lower semicontinuous, hence Borel, for the weak topology on probability laws. If  $\mathcal{G}_L \uparrow \mathcal{G}$  generates the full sigma-field, then*

$$D_{\text{KL}}(P|_{\mathcal{G}_L} \| Q|_{\mathcal{G}_L}) \uparrow D_{\text{KL}}(P\|Q). \quad (78)$$

Consequently, if  $Z_L$  are increasing prefixes generating a complete transcript  $Z$ , then  $I(A; Z_L) \uparrow I(A; Z)$ .

*Proof.* The variational formula

$$D_{\text{KL}}(P\|Q) = \sup_{\phi} \left\{ \int \phi dP - \log \int e^{\phi} dQ \right\}$$

over bounded continuous  $\phi$  proves lower semicontinuity, because each displayed functional is continuous in  $(P, Q)$ ; Borel measurability follows. Applying the same formula to bounded functions measurable with respect to  $\mathcal{G}_L$ , and then using martingale approximation of bounded  $\mathcal{G}$ -measurable functions, proves (78). For mutual information take the two laws to be  $P_{A,Z}$  and  $P_A \otimes P_Z$  and restrict both to  $\sigma(A, Z_L)$ .  $\square$

**Lemma C.3** (Finite-horizon selection for Gaussian BCO). *Let  $C$  be compact and let  $\mathfrak{F}_d(C)$  be a Borel subset of  $\mathcal{C}(C, [0, 1])$  under the uniform topology. Fix a Borel target kernel into a standard Borel space  $\mathbf{A}$  and a query cap  $L < \infty$ . The capped statewise value*

$$v_L(s) = \inf_{\mathcal{E}: N_Z \leq L} \rho(s, \mathcal{E}) \quad (79)$$

*is universally measurable. For every  $\epsilon > 0$ , there is a universally measurable state-to-experiment rule  $\sigma_{L,\epsilon}$  satisfying*

$$\rho(s, \sigma_{L,\epsilon}(s)) \leq v_L(s) + \epsilon \quad \text{at every active state } s. \quad (80)$$

*Proof.* The hidden variable for one scale is  $(F, A)$ , which takes values in a standard Borel space. Its joint posterior  $\mu$  is therefore a point of the standard Borel belief space  $\mathcal{P}(\mathfrak{F}_d(C) \times \mathbf{A})$ . Define

$$\ell(\mu, x) = \int \{f(x) - \min_C f\} \mu(df, da), \quad r(\mu) = \min_{x \in C} \ell(\mu, x).$$

If the initial state is  $s = (Q, b)$  and the target channel is  $K_b(da | f)$ , then

$$\mu_s(df, da) = Q(df)K_b(da | f).$$

The map  $s \mapsto \mu_s$  is Borel by measurable parameter integration. The integrand is bounded and jointly continuous in  $(f, x)$ , so  $\ell$  is Borel and continuous in  $x$ ; compactness of  $C$  makes  $r$  Borel and gives a Borel Bayes-action selector. The Gaussian observation kernel is Borel, and its strictly positive density gives the explicit Borel update

$$\Psi(\mu, x, o)(df, da) = \frac{\varphi(o - f(x)) \mu(df, da)}{\int \varphi(o - f'(x)) \mu(df', da')},$$

where  $\varphi$  is the standard Gaussian density. By Lemma C.2,

$$i(\mu, x) = I_\mu(A; O \mid X = x)$$

is Borel. It is finite uniformly in  $(\mu, x)$ : data processing and the Gaussian capacity bound give  $i(\mu, x) \leq I(F(x); F(x) + \xi) \leq \frac{1}{2} \log(5/4)$ .

For fixed  $\lambda > 0$ , the depth- $k$  offset problem has the belief-state recursion

$$\begin{aligned} J_k^\lambda(\mu, b) = \inf_{x \in C} & \left\{ b^2 + b\ell(\mu, x) - \lambda i(\mu, x) \right. \\ & \left. + \int [-\lambda d \mathbf{1}\{r(\mu') \leq \gamma b\} + \mathbf{1}\{r(\mu') > \gamma b\} W_{k-1}^\lambda(\mu', b)] P_\mu(do \mid x) \right\}, \quad (81) \\ W_0^\lambda = 0, \quad W_k^\lambda = \min\{0, J_k^\lambda\}, \quad \mu' = \Psi(\mu, x, o). \end{aligned}$$

The root uses  $J_L^\lambda$ , rather than  $W_L^\lambda$ , because an experiment has positive length;  $W_k^\lambda$  permits voluntary stopping after the first query. The posterior belief is a sufficient controlled state, so any capped history-dependent policy can be Markovized without changing its expected offset value (equivalently, the relevant belief-action occupancy measures). If  $\tau$  is its stopping time, the stopped information chain rule reads

$$I(A; Z) = \mathbb{E} \sum_{t=1}^{\tau} i(\mu_{t-1}, X_t). \quad (82)$$

Together these facts show by induction that  $J_L^\lambda$  equals

$$\inf_{\mathcal{E}: N_Z \leq L} \{c_s(\mathcal{E}) - \lambda g_s(\mathcal{E})\}, \quad c_s = b^2 \mathbb{E}_s N_Z + b \mathbb{E}_s R_Z, \quad g_s = I_s(A; Z) + d \mathbb{P}_s(\mathbf{M}). \quad (83)$$

Randomization causes no difficulty: its complete seed is independent of  $(F, A)$  given the state and is included in the transcript, so the offset is the average of the offsets of its deterministic realizations.

All primitive kernels and costs above are Borel, and the only negative one-step term is finite because  $i(\mu, x) \leq \frac{1}{2} \log(5/4)$ . For fixed  $L$  and  $\lambda$  the stage cost can therefore be shifted to be nonnegative before applying the standard integration theorem. The recursion preserves lower semianalyticity jointly in  $(\mu, b, \lambda)$ , and the measurable selection theorem gives universally measurable near-minimizers at every stage, with stagewise errors allocated to sum to any prescribed total (Bertsekas and Shreve, 1978, Chapter 7). Finally, for rational  $q > 0$ ,

$$v_L(s) < q \iff J_L^q(\mu, b) < 0. \quad (84)$$

The right-hand side is universally measurable, so the rational strict sublevel sets in (84) first show that  $v_L$  is universally measurable. To construct the policy, enumerate  $\mathbb{Q}_{>0} = \{q_1, q_2, \dots\}$ . On  $\{v_L < \infty\}$ , choose measurably the first index  $j(s)$  such that

$$v_L(s) < q_{j(s)} \leq v_L(s) + \epsilon/2,$$

and then the first  $m(s)$  satisfying

$$\frac{1}{m(s)} < -\frac{1}{2}J_L^{q_{j(s)}}(\mu_s, b).$$

For each pair  $(j, m)$ , the dynamic program has a universally measurable  $1/m$ -optimal offset policy. Patch these countably many policies on the sets  $\{j(s) = j, m(s) = m\}$ . Its induced cost and progress satisfy

$$c_s - q_{j(s)}g_s \leq J_L^{q_{j(s)}}(\mu_s, b) + \frac{1}{m(s)} < 0.$$

Thus  $g_s > 0$  and  $\rho(s, \sigma_{L, \epsilon}(s)) < q_{j(s)} \leq v_L(s) + \epsilon/2$ . On  $\{v_L = \infty\}$ , any measurable one-query rule suffices and the desired inequality is vacuous. The patched stagewise rules form one universally measurable block policy on the augmented state, proving (80).  $\square$

**Proposition C.4** (Selection and truncation). *Let  $\mathbf{S}$  be a standard Borel state space, let  $\mathcal{A}_L(s)$  be the admissible first-hit experiments using at most  $L$  queries, and let  $v_L(s)$  be their statewise infimum. Suppose that every  $v_L$  has the measurable selectors in (80). Then the unrestricted value*

$$v(s) = \inf_{\mathcal{E} \in \mathcal{A}(s)} \rho(s, \mathcal{E}) \quad (85)$$

*is universally measurable and has a universally measurable  $\epsilon$ -optimal selector for every  $\epsilon > 0$ . The selected experiment may be taken to have a finite deterministic cap  $L(s)$  depending on the state. Consequently the statewise and measurable-policy profiles coincide.*

*Proof.* For an admissible experiment  $\mathcal{E}$ , stop it after at most  $L$  queries and denote the resulting first-hit experiment by  $\mathcal{E}^{[L]}$ . Encode each truncation as a nested transcript prefix with a stopped marker and dummy padding, so that  $\sigma(Z^{[L]}) \uparrow \sigma(Z)$ . Since regret is nonnegative and  $\mathbb{E}N_Z < \infty$ ,

$$c_s(\mathcal{E}^{[L]}) \uparrow c_s(\mathcal{E}), \quad \mathbb{P}_s(\mathbf{M}^{[L]}) \uparrow \mathbb{P}_s(\mathbf{M}).$$

The prefix sigma-fields increase to that of the complete almost surely finite transcript, so Lemma C.2 gives  $I_s(A; Z^{[L]}) \uparrow I_s(A; Z)$ . Thus  $g_s(\mathcal{E}^{[L]}) \uparrow g_s(\mathcal{E})$  and

$$\rho(s, \mathcal{E}^{[L]}) \longrightarrow \rho(s, \mathcal{E}), \quad (86)$$

including the zero-denominator case under the value  $+\infty$  convention. It follows that  $v = \inf_{L \geq 1} v_L$ , hence  $v$  is universally measurable.

Choose measurably

$$L_\epsilon(s) = \min\{L : v_L(s) \leq v(s) + \epsilon/2\}$$

and, on the universally measurable set  $\{s : L_\epsilon(s) = L\}$ , use  $\sigma_{L, \epsilon/2}$ . The patched rule stores  $L_\epsilon(s)$  at block entry and is one universally measurable policy on the augmented state. It has ratio at most  $v(s) + \epsilon$  at every state. Taking the supremum over states and then  $\epsilon \downarrow 0$  proves equality of the profiles.  $\square$

*Proof of Theorem 2.3.* In Gaussian BCO, Lemma C.3 supplies a universally measurable near-minimizer under every finite cap. Proposition C.4 then supplies one measurable state-dependent experiment for the uncapped value, proving (25).  $\square$

*Proof of Theorem 2.4.* Theorem 2.3 identifies the two profiles and hence their polynomial growth exponents. The exponent sandwich in Theorem 2.1 identifies the Bayes exponent, while Proposition 2.2 identifies the Bayes and minimax exponents. This proves (26).  $\square$

*Proof of Proposition 2.5.* The dense linear prior in Proposition A.8 gives the lower bound in (27); the general BCO regret bound and Proposition A.11 give the upper bound.  $\square$

## D Proof of the observation-adaptive dimple experiment

Throughout this appendix fix a target  $0 < a \leq b_0$  and total failure probability  $\delta = c_\delta a$ , where  $c_\delta > 0$  will be chosen after  $\gamma_0 < \gamma$ , and put  $\Lambda = 1 + \log\{16d(1 + \log(d/a))/\delta\}$ . Constants below are universal and may change from line to line. We use repeatedly the standard consequence of sphere concentration that, for a Haar orthonormal basis  $(u_i)$  in a fixed  $k$ -dimensional subspace and fixed unit  $w$ ,

$$\max_i |\langle w, u_i \rangle| \leq C\sqrt{\Lambda/k} \quad (87)$$

outside probability  $e^{-c\Lambda}$  when  $\Lambda \leq k$ ; when  $\Lambda > k$ , we use the deterministic bound 1. All repetitions below use fresh Gaussian noise. Each stage has a deterministic query cap, the number of stages is at most  $C(1 + \log(d/a))$ , and confidence levels are divided across these caps so that the union of all concentration failures is at most  $\delta$ .

Every use of Haar concentration is conditional on the transcript before the corresponding basis is drawn. At that point the current iterate, the hidden direction, and the activity margin are fixed, while the new basis is Haar in the stated tangent space. Thus the displayed tail bound applies conditionally at each stage. Iterated conditioning and the allocated union bound give one simultaneous event on which all activity, estimation, and contraction invariants below hold. The deterministic caps define the routine on its complement, so no assertion relies on an indefinitely repeated success event.

Whenever  $r = \|\theta - x\| < s$ , expanding (28) gives the exact identity

$$F_\theta(x) = s\|\theta - x\| + \langle \theta, x \rangle - 1. \quad (88)$$

**Lemma D.1** (Orthogonal initialization). *With probability at least  $1 - \delta/3$ , the following construction returns a unit vector  $v_0$  satisfying*

$$\langle \theta, v_0 \rangle \geq c \min \left\{ 1, \max \left( \sqrt{\varepsilon}, \sqrt{\Lambda/d} \right) \right\}, \quad (89)$$

*using at most  $Cd^2\Lambda^2/\varepsilon^2$  queries and  $Cd^2\Lambda^2/\varepsilon$  regret on the joint success event. If  $D$  is the noiseless centered-difference vector and  $\zeta$  is its Gaussian estimation error, then*

$$D = -b_0\theta + e_0, \quad \|e_0\| \leq Ch_0^2, \quad |\langle \theta, \zeta \rangle| \leq \frac{C\varepsilon}{h_0\sqrt{d\Lambda}}, \quad \|\zeta\| \leq \frac{C\varepsilon}{h_0\Lambda} \left( 1 + \sqrt{\frac{\Lambda}{d}} \right). \quad (90)$$

*Proof.* Draw a Haar basis  $(u_i)_{i=1}^d$ , set

$$h_0 = c_0 \min\{\sqrt{\varepsilon}, \varepsilon\sqrt{d/\Lambda}\}, \quad n_0 = \lceil Cd\Lambda^2/\varepsilon^2 \rceil, \quad (91)$$

and estimate every centered difference

$$D_i^0 = \frac{F_\theta(h_0 u_i) - F_\theta(-h_0 u_i)}{2h_0}$$

using  $n_0$  repetitions at each of the two points. On (87),

$$h_0^2 + 2h_0 \max_i |\langle \theta, u_i \rangle| < 2\varepsilon$$

for small enough  $c_0$ , so both points lie in the active dimple. Directly from (88),

$$D_i^0 = -\lambda_i \langle \theta, u_i \rangle, \quad \lambda_i = \frac{2s}{r_i^+ + r_i^-} - 1, \quad |\lambda_i - b_0| \leq Ch_0^2, \quad (92)$$

where  $r_i^\pm = \|\theta \mp h_0 u_i\|$ . Hence the mean difference vector is  $-b_0\theta + e_0$  with  $\|e_0\| \leq Ch_0^2$ .

The Gaussian error vector  $\zeta$  is isotropic with coordinate variance  $(2n_0h_0^2)^{-1}$ . At the allocated joint failure level,

$$|\langle \theta, \zeta \rangle| \leq \frac{C\varepsilon}{h_0\sqrt{d\Lambda}},$$

$$\|\zeta\| \leq \begin{cases} C\varepsilon/(h_0\Lambda), & \Lambda \leq d, \\ C\varepsilon/(h_0\sqrt{d\Lambda}), & \Lambda > d. \end{cases}$$

Let  $t = \min\{1, \max(\sqrt{\varepsilon}, \sqrt{\Lambda/d})\}$ . When  $\Lambda \leq d$ , the definition of  $h_0$  gives  $h_0 \asymp c_0\varepsilon/t$ . Hence, after first choosing  $c_0$  small and then the repetition constant large,

$$\langle \theta, -\hat{D}^0 \rangle \geq b_0 - Ch_0^2 - |\langle \theta, \zeta \rangle| \geq c\varepsilon, \quad \|\hat{D}^0\| \leq C\varepsilon/t.$$

When  $\Lambda > d$ , one has  $t = 1$ ; the second line of the noise bound,  $\varepsilon \geq d^{-1}$ , and the definition of  $h_0$  instead give the same two inequalities with  $t = 1$ . Therefore

$$v_0 = -\hat{D}^0/\|\hat{D}^0\|, \quad \langle \theta, v_0 \rangle \geq \min\{c_\star, c_1 \max(\sqrt{\varepsilon}, \sqrt{\Lambda/d})\} =: c_{\text{init}}. \quad (93)$$

Here  $c_\star > 0$  is a small universal constant. This stage uses  $\tilde{O}(d^2/\varepsilon^2)$  queries. Its query points have loss  $O(\varepsilon)$ , so its regret is  $\tilde{O}(d^2/\varepsilon)$ ; these are bounded by  $\tilde{O}(d^2/a^2)$  and  $\tilde{O}(d^2/a)$  because  $a \leq b_0 \asymp \varepsilon$ .  $\square$

**Lemma D.2** (Tangent amplification). *Conditionally on the initialization event, the capped routine in this stage reaches  $\|\theta - v\| \leq q_\star$  with failure probability at most  $\delta/3$ , using*

$$N_{\text{amp}} \leq \frac{Cd^2\Lambda}{c_{\text{init}}^4}, \quad R_{\text{amp}} \leq \frac{Cd^2\Lambda}{c_{\text{init}}^2}, \quad (94)$$

*up to lower-order  $C\Lambda(1+\log(1/c_{\text{init}}))$  acceptance-test costs. At every continuing stage with correlation at least  $c$ , its probes are active, its noiseless tangent estimate  $g$  and Gaussian error  $\zeta$  satisfy*

$$\langle \theta, g \rangle \geq c_1c^2, \quad \|g\| \leq Cc, \quad |\langle \theta, \zeta \rangle| \leq \frac{1}{2}c_1c^2, \quad \|\zeta\| \leq Cc, \quad (95)$$

*and the new correlation lower bound grows by a fixed factor.*

*Proof.* Fix  $q_\star$  first. Suppose a unit vector  $v$  satisfies  $\langle \theta, v \rangle \geq c$ , where  $c_{\text{init}} \leq c \leq 1 - q_\star^2/8$ . All activity and Taylor estimates below are uniform on this interval. In particular,  $c^2 \geq c_2\varepsilon$  after changing universal constants. On the unit sphere the value is the nondecreasing function

$$\phi_\varepsilon(q) = \varepsilon + \frac{1}{2} - \frac{1}{2}(s - q)_+^2, \quad q = \|\theta - v\|.$$

Choose  $q_\star > 0$  so small that  $q_\star < s/4$ . Uniformly over  $0 < \varepsilon \leq \varepsilon_0$ , the gap

$$\Delta_\star = \phi_\varepsilon(q_\star) - \phi_\varepsilon(q_\star/2)$$

is bounded below by a positive universal constant. Compare the empirical value at  $v$  with the midpoint of these two values. Gaussian concentration with  $O(\Lambda/\Delta_\star^2) = O(\Lambda)$  repetitions separates  $q \leq q_\star/2$  from  $q \geq q_\star$ ; on the overlap it may take either branch. Thus stopping implies  $q \leq q_\star$ , while continuing implies  $q \geq q_\star/2$ . On the active part the same value is the strictly increasing function

$$\psi(q) = sq - q^2/2, \quad (96)$$

for a fixed sufficiently small  $q_\star > 0$ .

On the latter branch, put  $x = (c/2)v$  and draw a Haar tangent basis  $(u_i)_{i=1}^{d-1}$  of  $v^\perp$ . When  $\Lambda \leq d$ , set  $h = c_2 c$ ; by (87) and  $c \geq c_{\text{init}}$ , all  $x \pm hu_i$  remain in the active dimple. On the capped branch  $\Lambda > d$ , set  $h = c_2 c^2$  instead. Since  $\|\theta - x\|^2 \leq 1 - 3c^2/4$ , this choice is deterministically active for small  $c_2$ ; here  $c \geq \min\{c_\star, c_1\} =: c_{\text{low}} > 0$ , and changing  $h$  only changes universal constants in the repetition budget. If  $q_i = \langle \theta, u_i \rangle$ , their exact central differences satisfy

$$D_i = -\lambda_i q_i, \quad \lambda_i = \frac{2s}{r_i^+ + r_i^-} - 1. \quad (97)$$

Writing  $r_0 = \|\theta - x\|$  and  $\lambda_0 = s/r_0 - 1$ , elementary Taylor bounds on the square root, uniformly for  $r_i^\pm \geq 1/2$ , give

$$\lambda_0 \geq c_3 c^2, \quad \lambda_0 \leq Cc, \quad |\lambda_i - \lambda_0| \leq Ch^2. \quad (98)$$

Put

$$g = -\sum_i D_i u_i = \sum_i \lambda_i q_i u_i, \quad w = P_{v^\perp} \theta = \sum_i q_i u_i.$$

On the continuing branch,  $\|\theta - v\| \geq q_\star/2$ , while the correlation lower bound and the choice of  $k$  below keep  $\langle \theta, v \rangle$  positive. Thus  $\|w\|$  is bounded below by a positive constant depending only on  $q_\star$ . Equations (98) and  $h \leq c_2 c$  give, for sufficiently small  $c_2$ ,

$$\langle \theta, g \rangle = \sum_i \lambda_i q_i^2 \geq c_4 c^2, \quad \|g\|^2 = \sum_i \lambda_i^2 q_i^2 \leq Cc^2.$$

Estimate each  $D_i$  with  $n_c = \lceil Cd\Lambda/c^4 \rceil$  repetitions at each side. Choosing the repetition constant uniformly over the capped stages, Gaussian concentration in  $v^\perp$  and the lower bound  $c \geq c_{\text{init}} \gtrsim \sqrt{\Lambda/d}$  (in the nontrivial  $\Lambda \leq d$  case) make the error  $\zeta$  satisfy

$$\|\zeta\| \leq c_5 c, \quad |\langle \theta, \zeta \rangle| \leq (c_4/2)c^2$$

on the joint event. Thus the normalized negative estimate  $u \in v^\perp$  obeys

$$\langle \theta, u \rangle \geq k_0 c \quad (99)$$

for a universal  $k_0 > 0$ . Choose  $0 < k \leq \min\{k_0, q_\star/4\}$  once and for all. Replacing  $v$  by  $(v + ku)/\sqrt{1+k^2}$  raises its correlation lower bound from  $c$  to at least  $\sqrt{1+k^2}c$ ; on the continuing branch the choice of  $k$  keeps this lower bound at most one. Iterate until it exceeds  $1 - q_\star^2/8$ . At that point  $\|\theta - v\| \leq q_\star/2$ , so the radial gate must stop. There are at most  $C \log(2/c_{\text{init}}) = O(\log d)$  stages. The stage at correlation  $c$  uses  $O(d^2\Lambda/c^4)$  queries. Moreover  $F_\theta(x \pm hu_i) \leq C(\varepsilon + c^2) \leq Cc^2$ , so geometric summation from  $c_{\text{init}}$  gives

$$N_{\text{amp}} = \tilde{O}(d^2/\varepsilon^2), \quad R_{\text{amp}} = \tilde{O}(d^2/\varepsilon). \quad (100)$$

□

**Lemma D.3** (Angular contraction). *Conditionally on the first two stage events, this capped stage returns  $\hat{x}$  with  $F_\theta(\hat{x}) \leq \gamma_0 a$  outside probability  $\delta/3$ , and has deterministic query cap  $Cd^2\Lambda/a^2$  and success-event regret at most  $Cd^2\Lambda/a$ . At any intermediate scale  $\|\theta - v\| \leq Q \leq q_\star$ , one step uses at most  $Cd^2\Lambda/Q^2$  queries, incurs at most  $Cd^2\Lambda/Q$  regret on its success event, and returns  $v_{\text{new}}$  with  $\|\theta - v_{\text{new}}\| \leq Q/2$ .*

*Proof.* Now  $q = \|\theta - v\| \leq q_*$ . Maintain the deterministic invariant  $\|\theta - v_j\| \leq Q_j$ , where  $Q_j = q_* 2^{-j}$ , and at one step write  $v = v_j$  and  $Q = Q_j$ . Take  $h = c_6 Q$  and draw a Haar tangent basis  $(u_i)$  of  $v^\perp$ . Define

$$\alpha_h = (1 + h^2)^{-1/2}, \quad t = h(1 + h^2)^{-1/2}, \quad x_i^\pm = \alpha_h v \pm t u_i.$$

All these points lie on the unit sphere and within distance  $CQ$  of  $\theta$ , hence in the active duple. Write

$$\theta = c_v v + \sum_i q_i u_i, \quad c_v = 1 - q^2/2, \quad G(p) = \psi(\sqrt{2 - 2p}).$$

Then exactly

$$F_\theta(x_i^\pm) = G(\alpha_h c_v \pm t q_i), \quad q_i = \frac{G^{-1}(F_\theta(x_i^+)) - G^{-1}(F_\theta(x_i^-))}{2t}. \quad (101)$$

Let

$$I_Q = \{p \in [-1, 1] : \sqrt{2 - 2p} \leq C_0 Q\}.$$

All inner products in (101) lie in  $I_Q$ . On this interval  $G$  is strictly monotone and, writing  $r = \sqrt{2 - 2p}$ ,

$$|(G^{-1})'(G(p))| = \frac{r}{s - r} \leq CQ.$$

Project each empirical function value onto  $G(I_Q)$  before applying  $G^{-1}$ . Estimating each value to accuracy  $c_7 Q/\sqrt{d}$ , using  $\lceil Cd\Lambda/Q^2 \rceil$  repetitions per site, changes each inverse inner product by at most  $Cc_7 Q^2/\sqrt{d}$ . Since  $t \asymp Q$ , every tangent coefficient changes by at most  $Cc_7 Q/\sqrt{d}$ . Consequently  $\|\hat{w} - w\| \leq Cc_7 Q$ , where  $w = \sum_i q_i u_i$  and  $\hat{w} = \sum_i \hat{q}_i u_i$ . Clip  $\hat{w}$  to norm at most  $1/2$  and set

$$v_{\text{new}} = \sqrt{1 - \|\hat{w}\|^2} v + \hat{w}.$$

On the ball  $\|w\| \leq 1/2$ , the map  $w \mapsto \sqrt{1 - \|w\|^2} v + w$  is 2-Lipschitz. Choose  $q_*$  and then  $c_7$  sufficiently small. Projection onto the radius-1/2 ball is nonexpansive, so  $\|v_{\text{new}} - \theta\| \leq 2Cc_7 Q \leq Q/2$ .

At angular scale  $Q$ , the query count and regret are

$$N_Q = \tilde{O}(d^2/Q^2), \quad R_Q = \tilde{O}(d^2/Q), \quad (102)$$

because every probe has loss  $\psi(O(Q)) = O(Q)$ . Summing geometrically until  $Q \leq c_5 a$  proves the query and regret bounds in (71). At termination,  $F_\theta(\hat{x}) = \psi(\|\theta - \hat{x}\|) \leq \gamma_0 a$ .  $\square$

*Proof of Theorem B.2.* Run Lemmas D.1–D.3 in order, assigning failure probability  $\delta/3$  to each. Their deterministic caps and geometric sums give  $N_Z \leq \tilde{O}(d^2/a^2)$  and success-event regret  $\tilde{O}(d^2/a)$ .

On the complement of the joint concentration event the capped routine uses no more than its displayed query budget and incurs at most one unit of regret per query. Since its probability is at most  $\delta = c_\delta a$ , this adds only  $\tilde{O}(d^2/a)$  expected regret. The frequentist guarantee is uniform in  $\theta$ , so under every prior

$$\mathbb{E}[F_\Theta(\hat{x})] \leq (\gamma_0 + c_\delta) a =: a_0 a.$$

Choose  $c_\delta$  so that  $a_0 < \gamma$  and apply Lemma A.3. It accepts with probability at least  $1 - a_0/\gamma > 0$  and gives the required terminal posterior-risk bound.  $\square$

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